University of Portsmouth
M32616 - 2024/25 - Research Project

# Investigating the Viability of Machine Learning Models for Risk Prediction with Datix & Supporting Data

Callum Aldridge
UP2302645
September 2025

**Abstract**

Violence and aggression in mental health inpatient settings implicitly pose risks to patient safety and therapeutic progress. Traditional risk assessment tools provide valuable insight; however, they are limited in responsiveness and adaptability to dynamically changing patient behaviour. In collaboration between the University of Portsmouth and St Andrew's Healthcare, anonymised Datix incident reports and supporting patient data are used to train and evaluate predictive models. The investigation intends to assess whether there is potential to incorporate such models into clinical workflows, for the purpose of supporting proactive interventions and optimising care in secure and forensic environments, in the future. The desired outcome is to produce further supporting evidence for the applicable and effective use of machine learning models in this sector.

# Contents

**1 Introduction** **3**
**2 Literature Review** **5**
2.1 Similar Studies 5
2.1.1 Machine-learning prediction of mental-health crises from electronic health records (Garriga et al, 2022) 5
2.1.2 Data-science prediction of aggressive incidents in psychiatric inpatients (Suchting et al, 2018) 5
2.2 Machine Learning 6
2.2.1 Deep Learning, Neural Networks & Temporal Models 6
2.2.2 Performance Metrics 7
2.2.3 Explainable Artificial Intelligence (XAI) 7
2.3 Clinical Relevance Papers 8
2.3.1 Violence Risk Assessments and Clinical Scales 8
2.3.2 Clinical Guidelines 9
2.4 Discussion and Relevance 9
**3 Methodology** **11**
3.1 Data Pre-Processing, Cleaning & Structuring 11
3.2 Recurrent Neural Networks (RNNs) 12
3.2.1 Rationale and Specific Structuring 12
3.2.2 Model Architecture 12
3.3 Transformers 13
3.4 Performance Metrics 14
**4 Results & Evaluation** **15**
4.1 Dataset Visualisations & Basic Statistics 15
4.2 Model Performance Statistics 24
4.2.1 RNN Architecture 24
4.3 XAI Technique Outputs 31
4.4 UI Widget 33
4.5 Transformer Tinkering 34
**5 Discussion** **36**
5.1 Machine Learning Performance Discussion 36
5.1.1 Model Comparison 36
5.2 Clinical Discussion 39
5.2.1 Clinical Insights 39
5.2.2 Clinical Implications 41
**6 Conclusion** **43**
**7 Acknowledgements** **44**
**8 References** **47**

9 Appendix 53

# 1 Introduction

Violence & Aggression with regard to mental health inpatients is a prevailing hazard for clinicians & patient peers. Violent incidents, although arguably inevitable, compromise general safety and disrupt the therapeutic environment. Further, the management of a violent incident can lead to adverse clinical and legal outcomes (National Institute for Health and Care Excellence, 2015). The established approach to prevention/mitigation of violent incidents relies on staff members' clinical judgement, structured risk assessment tools (such as HoNos, HCR-20 & Brøset) and historical behavioural information that is reviewed and deliberated upon via human inspection. These methods have unequivocally improved the understanding of patients with violent tendencies and risks over time. However, they are limited by time constraints, the constantly changing presentation of an individual and most importantly, the subjectivity of the assessor. Although with reference to existing risk assessments, proactive and individualised care, implicitly, can only be so thoroughly described by a discrete checklist.

Recent advances in Machine Learning and Artificial Intelligence provide a potential opportunity to complement existing clinical practice, should attention be brought to development in this field. Data-driven, real-time risk prediction systems present an opportunity to utilise the prolific font of data that is collected by healthcare providers annually and further utilise the complex and lateral pattern recognition of deep machine learning architectures.
Recent studies in this crossover of sectors have provided evidence in favour of potential effective and appropriate utilisation of these techniques for inpatient care within mental health services (Garriga et al, 2022; Suchting et al, 2018). Systems of this fashion have the potential to provide probabilistic forecasts within a clinically relevant timeframe in an effort to support clinicians with an early warning system for crisis events.

This project, conducted with support from the University of Portsmouth and St Andrews Healthcare, aims to investigate the viability of machine learning models for short-term violence risk prediction in inpatient mental health care. The study utilises anonymised Datix incident reports as well as additional supporting patient data to train and evaluate models capable of predicting the probability of a violent incident within a 24-hour window. The intended outcome is to assess whether such a system could function as an early warning tool, support for proactive management and/or enhancement of patient care. Whilst further discussing the utilisation of the machine learning tools for this purpose, and whether the field of Artificial Intelligence & Machine Learning has advanced enough to provide sufficiently complex and appropriate architectures for predictive modelling of this kind.

By integrating structured incident and supporting characteristic data formatted into a time-series dataset, the investigation intends to further establish whether the gap between traditional clinical risk assessment tools and predictive analytics can be bridged. Optimistically, the findings may inform the development of future risk-aligned tools that are ethical, operationally viable and clinically meaningful within this setting.

# 2 Literature Review

## 2.1 Similar Studies

### 2.1.1 Machine-learning prediction of mental-health crises from electronic health records (Garriga et al, 2022)

This paper discusses the development of one of the first generalisable crisis prediction systems utilising electronic health records (EHR) that are routinely collected. The study highlights the general high demand for mental health (MH) services and goes on to comment on the impracticality of manually reviewing complex patient records. Therefore, justifying some experimentation in automated methods. Data from 17'122 patients from 2012 to 2018, they trained a gradient-boosting model to estimate the probability of a crisis episode within the next 28 days. The model achieved an AUROC score of 0.797 and an AUPRC score of 0.159; crises were predicted with 58% sensitivity and 85% specificity. Prospective deployment established that clinicians found the predictions valuable for caseload management and crisis mitigation in 64% of cases.

In most aspects, the paper illustrates how a machine learning model that utilises EHR is capable of identifying patterns in behaviour that precede mental health crises for a given patient. The dataset utilised in their investigation is stated to have contained satisfactory diversity across features such as 'age', 'diagnosis', etc. The study is in support of the feasibility of crisis prediction in this manner. Further, it establishes a reasonable benchmark for any relevant upcoming research that intends to develop risk predictions or early warning systems in this sector.

### 2.1.2 Data-science prediction of aggressive incidents in psychiatric inpatients (Suchting et al, 2018)

With data collected from 29'841 admissions in Houston, TX, from 2010 to 2015, this study also utilises routinely collected electronic health records. Overall, they found that aggressive incidents were minimal across the scope of their study. Data utilised for this investigation were collected from nursing & medical assessments, as well as psychosocial assessments & psychiatric examination. This investigation utilised random forest, gradient boosting, deep neural network & penalised generalised linear models for the purpose of creating an early

warning system. Across each model, they achieved an AUC in the range of 0.71 to 0.78. The best being the penalised GLM that further identified several socio-forensic risk factors. The strongest retained predictor by this model was homelessness. The investigation suggests that there is an increased likelihood of aggression [67.5%] should the patient have been homeless prior to admission. However, the models highlighted that those who perpetrated and/or witnessed abuse were less likely to be aggressive whilst in hospital in this case.

The final model was capable of moderately predicting crisis events and flagging aggressive individuals. 18 out of 84 patients were flagged in a test set at a 6% threshold, illustrating some challenges that are implied to be in data quality/processing.

The investigation showed that structured EHR data can be utilised to create short-term predictors for aggression. However, we see that the data used in this case focused more heavily on the characteristics of the individual and their forensic and personal history. They were able to identify a modest number of high-risk patients that may require enhanced support, but state they require external validation, given the research was conducted at a single institution over a moderate period of time. The focus on social and forensic variables in this investigation has helped illustrate a higher prevalence of aggression for inpatients who could be considered socioeconomically disadvantaged. Overall, the paper supports *1.1.2* in the use of data science for proactive risk prediction and management.

## 2.2 Machine Learning

2.2.1 Deep Learning, Neural Networks & Temporal Models

Deep Learning (Goodfellow et al, 2016) provides an introduction to neural network architectures and optimisation methods to establish foundational content knowledge. Long short-term memory (LSTM) networks address the ‘vanishing gradient’ problem seen in traditional Recurrent Neural Networks [SimpleRNNs]. This is achieved by incorporating additional gates and a memory cell in its infrastructure such that gradients can propagate across many time steps (Hochreiter & Schmidhuber, 1997). LSTMs are best suited to longer sequences of time in comparison to SimpleRNNs. Gated Recurrent Units (GRU) effectively combat the vanishing gradient problem and also manage to reduce the time complexity of the algorithm by using a simpler gating mechanism (Chung et al., 2014; Vaswani et al., 2017; Kingma & Ba, 2014). Recurrent Neural Networks using these architectures appear provisionally appropriate for the investigation, given their temporal disposition and that the subject dataset is formatted as time series data.

This investigation also proposes the use of Transformer Architecture to contrast with the traditional temporal machine learning algorithms. This approach relies on built-in attention

mechanisms rather than recurrence and tends to be more viable in this context over longer time sequences. The ability for parallel attention to consider all past events could allow the model to condition on long-range clinical histories more proficiently. (Vaswani et al, 2017)

### 2.2.2 Performance Metrics

Given that the investigation is striving for probability outputs, it would be prudent to assess the accuracy and calibration of those outputs. The Brier score appears appropriate given the real-world nature of the data and the hypothetical implications of incorrect predictions. This is because it measures the Mean Squared Difference (MSE) between the predicted and actual outcomes to score performance. Further, it can be discerned whether discrimination or calibration is responsible for the result (Murphy, 1973; Flach & Kull, 2015). Log loss (cross entropy) is another scoring rule that is sensitive to overconfident and incorrect predictions, which could prove valuable during the training of the models.

As well as the probability-focused metrics, discrimination metrics such as precision-recall area under the curve (PR-AUC) and AUROC are capable of evaluating without a fixed decision threshold. AUPRC is particularly relevant when events are rare, as it focuses on the performance in the positive class, whereas ROC-AUC can appear overly optimistic under severe imbalance (Flach & Kull, 2015). These metrics were also utilised by the similar studies described in *2.1* and allow for direct comparison of performance.

Additionally, F1 score was used to capture performance at a given threshold, balancing both precision and recall. This is valuable in this context as it reflects how well models identify true violent incidents while limiting false alarms ensure that some sensitivity is achieved with excess in either false alarms or missed alarms.

### 2.2.3 Explainable Artificial Intelligence (XAI)

Hassija et al. (2023) review explainable AI methods and discuss some of the fundamentals of this area of research. They spend time discussing how interpretability is critical in certain sectors such as healthcare. SHAP and LIME are some of the core methods utilised to interpret model outputs and establish transparency and trust for black box models (Hassija et al., 2023). Palácios et al. (2021) provide a unified framework for XAI and argue for integrating explainability into clinical decision support to ensure models are created and used responsibly. There is further elaboration on the effective use of XAI methods as they can provide insight into the most influential features of individual predictions (Salih et al., 2023). Should any kind of model inform risk management decisions, interpretability and explainability are paramount to prevent bias, stigma or unjustified restriction on mental health service users.

## 2.3 Clinical Relevance Papers

2.3.1 Violence Risk Assessments and Clinical Scales

The Brøset Violence Checklist (BVC) is a tool used to predict violent behaviour within the next 24 hours (Almvik et al., 2000). It assesses confusion, irritability, boisterousness, verbal threats, physical threats and attacking objects, scoring each as present or absent. A BVC score greater than 2 implies a likelihood of imminent violence (Woods & Almvik, 2002). The BVC enables dynamic risk assessment and has been incorporated into some psychiatric training packages. The study suggests that dynamic patient behaviours, rather than static characteristics, are key determinants of violence in the near future (Woods & Almvik, 2002).

The 'Historical Clinical Risk Management 20' (HCR-20) tool comprises ten historical, five clinical and five risk management items and is widely considered the working standard for risk assessments, especially in forensic services. NHS England mandates completion of the HCR-20 for forensic patients every six months (Douglas & Webster, 1999; James et al., 2018; Palmstierna & Wistedt, 1987; Sugarman et al., 2007). Some have noted that structured tools for healthcare professionals to use, such as the HCR-20, could produce a false sense of accuracy and are typically time-consuming. However, the widespread use of the HCR-20 suggests a consensus of utility in practice, be that an actual or perceived utility (Douglas & Webster, 1999; James et al., 2018; Palmstierna & Wistedt, 1987; Sugarman et al., 2007). Overall, the tool informs care plans by considering past violence, present clinical presentation and potential future risk management needs.

The 'Dynamic Appraisal of Situational Aggression [Inpatient Version]' (DASA-IV) is another structured risk assessment tool, referenced by name in the NICE Guidelines [*see 2.3.2*], for crisis and violence prevention for psychiatric inpatients. It utilises seven items that are scored by the relevant clinician who evaluates the likelihood of an aggressive incident in the next 24 hours. This is based on recent behaviour such as irritability, impulsivity and verbal threats. Each of the seven items is scored on a scale of presence or absence preceding the assessment. The cumulative score produces a risk category and informs proportionate interventions (Daffern & Ogloff, 2015).

The 'Health of the Nation Outcome Scales' (HoNOS) is a clinician-rated measure used to assess psychiatric symptoms, social & functioning issues, as well as further guide/assist with care plans (Douglas & Webster, 1999; James et al., 2018; Palmstierna & Wistedt, 1987; Sugarman et al., 2007). For over two decades, the scales have been in use but recently have been updated to mitigate ambiguity and better reflect modern clinical practice (Douglas & Webster, 1999; James et al., 2018; Palmstierna & Wistedt, 1987; Sugarman et al., 2007). HoNOS is mandated for outcome monitoring in England. An iteration upon HoNOS, HoNOS-secure, incorporates seven security scales (A-G) for use in evaluating the level of

security required. This is most relevant to our investigation, given that HoNOS-secure quantifies an individual's needs and risks whilst in secure services. Each scale is scored from no risk to severe risk.

2.3.2 Clinical Guidelines

The National Institute for Health and Care Excellence (NICE) provides some evidence-based guidance for the prevention and management of violence and aggression in health and community settings. **NICE NG10** addresses the short-term management of violence and aggression in adults, young people and children for inpatients. The guidelines emphasise proactive prevention via recognition of early warning signs and structured risk assessment. They place special emphasis on the anticipation and de-escalation as a first response. Further, the NICE guidelines advise that restrictive interventions such as restraint, rapid tranquillisation or seclusion should only be used when other less restrictive approaches have failed and can be justified as the least restrictive course of action. NG10 recommends the use of actuarial tools such as the BVC or DASA-IV[*see 2.3.1*] in conjunction with multidisciplinary clinical judgement in efforts to manage the risk of violence and aggression. This, however, differs from guidance for self-harm and suicide. NICE guidance recommends a collaborative formulation approach instead of using risk stratification, in a fashion similar to BVC & DASA-IV, as a collaborative and holistic approach is considered more appropriate for crisis events of this nature.

Further, **NICE QS154** sets out quality standards when managing violent and aggressive behaviours. Emphasising person-centred, individualised care as well as regular risk assessment review and ensuring interventions are always appropriate and as respectful of patient dignity as possible. Overall, the guidelines intend to promote a balanced approach prioritising prevention and de-escalation whilst establishing the expected attitude for the more restrictive interventions.

## 2.4 Discussion and Relevance

The emerging machine learning techniques offer a promising opportunity to investigate the potential for the prediction of mental health crises and inpatient aggression. The 2022 Nature Medicine study demonstrates that continuous monitoring of EHR data can predict crises weeks ahead, enabling proactive and less restrictive intervention (Garriga et al, 2022). However, predictive performance remains moderate at this stage, reflecting the difficulty of forecasting rare events. The 2018 data science study suggests that social factors such as homelessness and forensic history strongly influence aggression risk (Salih et al, 2023). Although the data utilised for this investigation does not extend past clinical factors, this suggestion will likely be insightful as our analysis develops. The overall development of predictive machine learning models in this sector presents an implicit risk of inequality and bias if not trained and

developed with interpretability & transparency as well as foresight, as presented, even minimally, in these similar investigations.

These prior investigations, not unlike this one, do not utilise unstructured clinical notes. There is a potential lack of context presented in each investigation. A proposal for the use of Natural Language Processing (NLP) to identify certain items from a service user's progress notes would likely be a necessary step in incorporating plaintext in this fashion. Further, a specialised Large Language Model for this purpose may be able to extract key information to hand over to a predictive model, such as the aforementioned investigations. However, this is beyond the scope of this investigation.

Furthermore, while Explainable AI techniques offer promise, there appears to be a lack of effective utilisation of them. This establishes a limit on clinical trust and accountability. Future research should incorporate interpretability and evaluate models prospectively across diverse populations. Ethical considerations, including data privacy and the potential for internal stigmatisation, must be considered and incorporated into the central model design. Finally, predictive tools should not replace but complement clinical judgement and evidence-based interventions such as those outlined in the NICE guidelines and tried & tested risk-assessment tools.

# 3 Methodology

## 3.1 Data Pre-Processing, Cleaning & Structuring

Supplied by St Andrews were three separate datasets: Incidents (and Demographics), Medication History & Meaningful Activities. After cleaning and alignment, we constructed a patient-day timeline and further created arrays (X_padded, y_padded) due to the variance in sequence length because of different admission durations per patient. The consolidated dataset comprised 204’780 entries.

All relevant timestamps were parsed utilising datetime within Python and floored to midnight to allow consistent compatibility by day. Further, missing values were managed by utilising explicit fills for categorical values such as ‘Unknown’ & ‘None’ rather than dropping values and losing granularity overall. The ‘time_since_last_activity’ variable was filled with ‘-1’ when the information was unavailable.

All tables were aligned and features could be grouped into categories, for example, Violence signals (aggression_today, y_next24h), medication exposure (antipsychotic_24h, num_prn_72h, etc.), meaningful activities (activity_today, meaningful_activity_hours, etc.), safeguarding and observations (any_safeguarding, any_obs_event, any_selfharm), demographics (age, gender, etc.).

We initially intended to include correction of implausible values, negative or out-of-range values for any given features, but this proved unnecessary after inspection. At this juncture, it is

important to note that the St. Andrews Data Extraction team actively improved the data quality in the process of their extraction, thus mitigating the need for extended cleaning of this nature.

We constructed the sequences by day and produced a variable ‘y_next24h’ defined as a next-day shift of ‘aggression_today’ for a patient [groupby( ).shift(-1)]. The Recurrent Neural Network in this investigation utilises this feature for its predictions.
The final ‘Time Series’ reformatted dataset consisted of all days between admission and discharge (or end of timeline in the dataset), featuring days where there were no incidents. To establish a consistent input for the RNN model, we padded all sequences to a maximum length of [4495]. The time series dataset had fourteen features; however, RNNs have the capacity to consider the entire sequence before producing a prediction. This means that in extreme cases, prediction outputs could be supported by 62’930 [14 x 4495] inputs. However, this varies in reality, as the masking layer ignores any padded values, ensuring the model trains on only real data.

## 3.2 Recurrent Neural Networks (RNNs)

3.2.1 Rationale and Specific Structuring
The dataset represents a temporal progression of events for each patient through their admission. Features such as incident history, medication history and meaningful activities each change over time. Given that standard feed-forward models inherently treat each entry separately, they struggle to capture these interdependencies. The nature of RNNs allows them to consider previous timesteps when making predictions - effectively, retaining a memory of previous events. This appeared appropriate as previous patterns in patient behaviour may inform future likelihood of risk. Although we utilise Gated Recurrent layers, rather than Long Short-Term Memory layers, which are less computationally expensive, we also downsampled the dataset due to the growing time complexities of training, making the process unfeasible.

Each patient admitted within the dataset was converted into a per-day time-series sequence by incident. The binary target variable y_next24h was created to indicate whether an aggressive event took place in the next 24 hours from their last entry. Sequences (admission duration) varied from patient to patient, so they were padded to a maximum length, and a masking layer was applied during training to ignore these additional values.

3.2.2 Model Architecture
The investigation utilised different combinations of layers in an effort to produce results. A ‘hybrid’ RNN was attempted but was held back by the inclusion of a SimpleRNN layer, which was initially intended to capture short-term dependencies more effectively.

The architecture that achieved the best results at the conclusion of the investigation consisted of homogenous [active sequencing] Gated Recurrent Unit (GRU) bidirectional layers.

We also utilised Dense layers and Dropout layers to transform outputs into a risk estimate and help mitigate overfitting. The final sigmoid output layer produces a probability score between 0 and 1 for violence and aggression in the next 24 hours.

During training, we utilised a cross-entropy loss function and the Adam optimiser (Kingma & Ba, 2014). Further, class-weighting was implemented to help with any further imbalance in the data. Splitting the data by patient also ensured that entries could not exist in the training and validation set concurrently. Finally, early stopping was utilised with a patience of 8 epochs in an effort to mitigate overfitting (Brownlee, 2020).

After an effective RNN model was produced, we attempted to introduce an attention mechanism layer that was defined manually (Song et al., 2017). The existing RNN captures all the temporal dependencies. However, not all timesteps in a patient's journey through hospital are as immediately informative with respect to their risk. The attention mechanism introduced allowed for balancing between the importance of recent events in comparison to more historic incidents. This appeared prudent to incorporate the importance of recent incidents, which are more representative of an individual's present mental state, without also completely disregarding historically recorded risk.

The attention mechanism scores the importance of each timestep using a tanh activation function. This is then normalised using a softmax function to form weights across the entire sequence, which sum to 1. Finally, there is a final context vector which functions as a weighted sum representation of timesteps. The attention layer was incorporated after GRU outputs but before the Dense layers. In overview, it acts like a feature selector in time, highlighting the critical events/days rather than treating all data in the sequence equally. The attention layer also helps mitigate any effect from the padded values spanning before the start of the patient timeline.

### 3.3 Transformers

We spent time on another architecture as an auxiliary to the main investigation: Transformers. Although contributions from this effort are not as great, they provide another avenue for temporal predictions in machine learning that can be discussed (Lim et al., 2019).

When training the transformer model, the dataset was pre-processed differently. Initially, it is converted into a per-patient, per-day binary series. Fixed-length sliding windows are

established; each training instance is the prior 180 days of the binary series as an input, and the model predicts the next 90 days as a daily probability sequence. Unlike the RNN, padding is unnecessary as the inputs and outputs are fixed. We downsampled the data and ensured the balance between incident days and non-incident days was appropriate. The model is a compact Transformer, meaning the inputs are projected linearly, then positional embeddings are added and lastly, multi-head attention. A sequence representation is produced that is mapped to ninety outputs with a sigmoid function for daily probabilities. The same performance metrics are measured as in the main investigation for ease of comparison. Early stopping was also utilised to help mitigate overfitting.

The results of this model were intended as a proof of feasibility for this architecture. This is because the necessary fine-tuning, time complexity and computational resources grew out of the scope for this investigation as it progressed.

### 3.4 Performance Metrics

The prevailing appropriate and informative metrics throughout the investigation were F1 score, AUC score & Brier score.

*F1 score* demonstrated its relevant utility in its illustration of recall, precision and general risk of false alarm. If we consider the environment in which a tool like this is expected to be used, recall should be prioritised as a metric if a perfect model is not attainable. Provided intentions are to mitigate risk and harm, a high recall score will predict risk eagerly, although it is at risk of false alarms. However, should stringent allocation of resources (i.e. economically motivated management/allocation of staff for enhanced observations) be the priority, prioritisation of the precision metric will be preferable, although the model may be slower to react to increases in risk. Ideally, F1 score, as a representation of the balance of these two metrics, should be one of the primary metrics if a tool like the investigation proposes were to be feasible in reality, where the cost of resources is a factor to be considered.

*Area Under the Receiver Operating Characteristic Curve (AUC-ROC)* describes the discriminatory power of the model without an established threshold. In essence, a description of how the model can separate low-risk and high-risk days. A higher AUC in our case illustrates a strong capacity of the model to rank order patients with respect to their risk. Further, our similar studies utilised and showcased AUC scores in measuring the performance of their models, and therefore, it provides common ground for comparison as a metric.

*Brier Score;* Whilst F1 score measures error tradeoffs and AUC quantifies ranking, the Brier score measures how well probability outputs are calibrated. To expand, whether an output of 0.7 corresponds to a 70% chance of violence in reality, statistically. This would make a tool such as this more actionable, given that we are able to quantify how closely the model aligns with reality. It is defined as the mean squared error between predicted probabilities and actual outcomes, ranging from 0 to 1, with a lower score being desirable.

Accuracy was also tracked, but given the imbalance observed in the dataset, predicting no incident each day would provide a high accuracy on this dataset, as that is the prevailing nature of most entries within it.

The combination of these three metrics addresses the balance between missed risks & false alarms (F1 score); ensure the model is capable of distinguishing between risky/non-risky periods (AUC-ROC), as well as establish meaningful and potentially actionable outputs (Brier Score).

# 4 Results & Evaluation

## 4.1 Dataset Visualisations & Basic Statistics

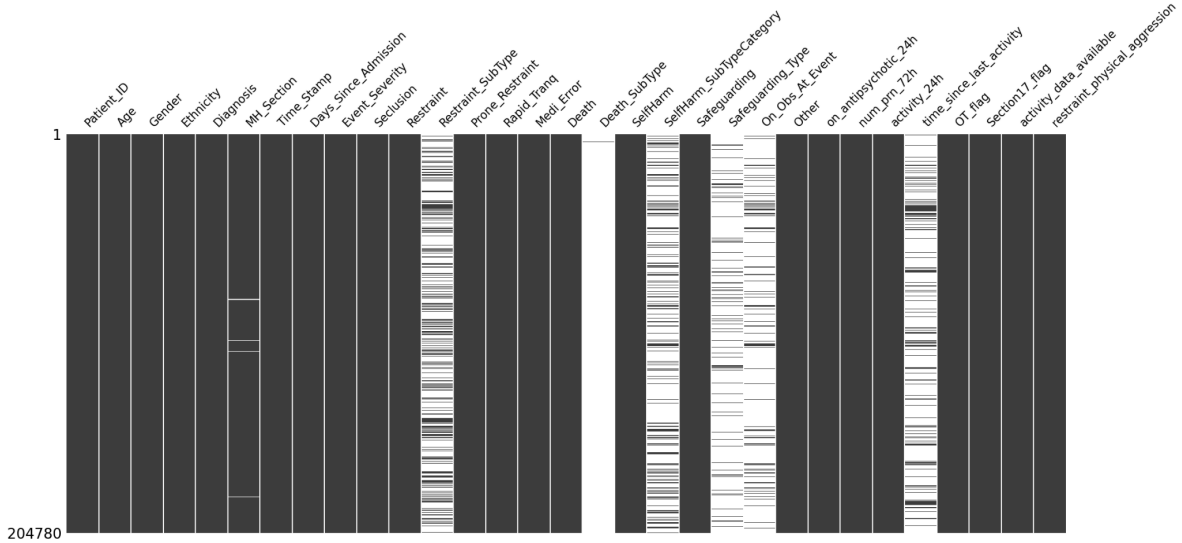



**Figure 1 - Missingness Map of Dataset**

*Figure 1* shows a 'missingness' map of the data. This is a visualisation that helps illustrate data quality through column-wise representation of missing/NaN values in the data, with white space marking empty values. Although we can observe six columns that largely contain missing information, upon further inspection, each of these belong to a 'sub_type' feature, i.e. further information about another feature. This implies that the dataset generally is not missing information but rather does not have some of the additional information that can be recorded when completing a Datix form.

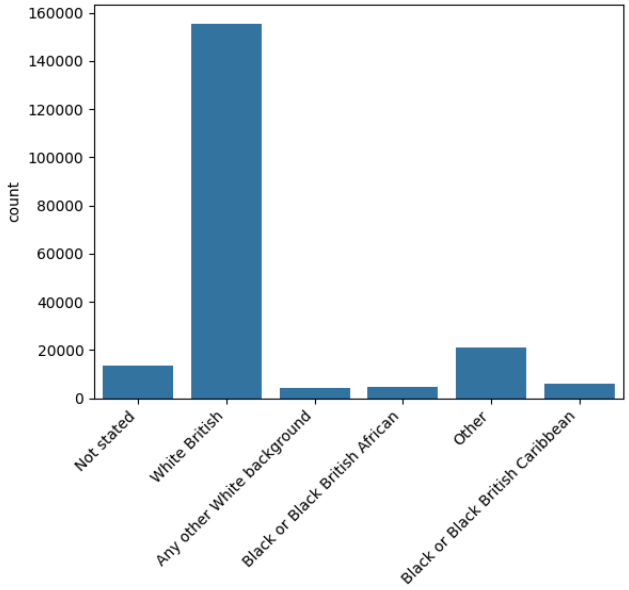



**Figure 2 - Distribution by Ethnicity**

*Figure 2* shows the distribution of patients by ethnicity represented in the dataset.
There is variation by ethnicity; however, the data is dominated by the White British demographic, creating an imbalance. The difference between the White British demographic and all ethnicities is unlikely to be mitigated completely with class weighting and other balancing techniques. In reality, training on this data will likely produce a model that produces less reliable predictions on minority groups.

St Andrew's Healthcare's largest site for inpatient mental health care is based in Northampton, UK. Although patients can be referred for private healthcare across the country, it stands to reason that the local community will still be some of the more common service users. A look at the UK Census (UK Census, 2021) suggests that the White British population is the most prevalent in the local area [85.9%]; there is some likelihood that this has been reflected in the dataset.

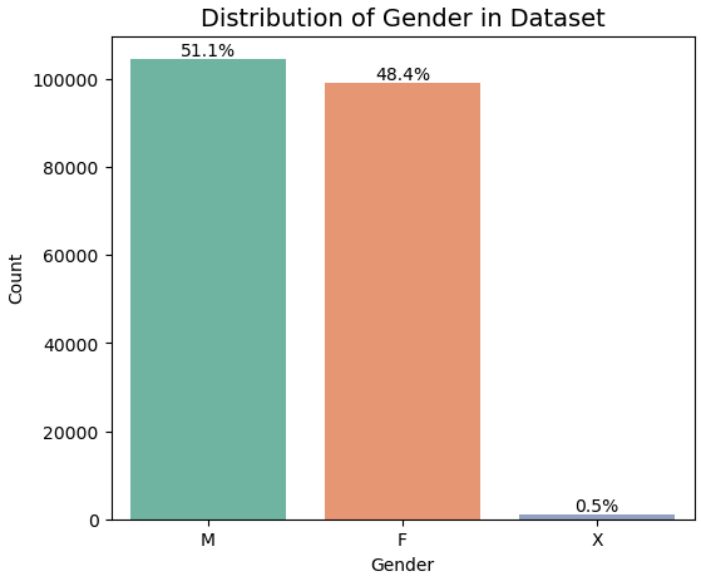



**Figure 3 - Distribution by Gender**

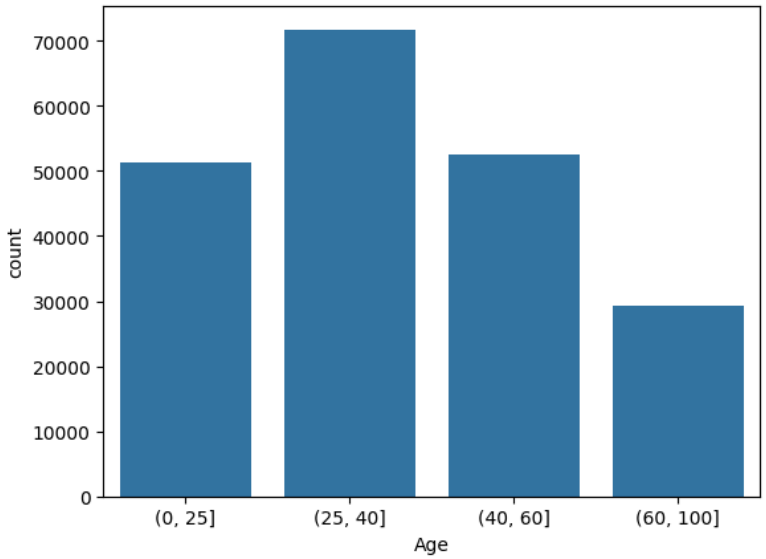



**Figure 4 - Distribution by Age (Binned)**

*Figures 3 & 4* illustrate the distribution of patients by gender and age. We see in *figure 3* that the dataset is reasonably balanced in this respect. *Figure 4* shows that ages are concentrated in the 25-40 year old band, working age adults. This distribution implies that prediction may be more unstable in the upper age bands, although this is not to imply that there is no representation for this group at all. Child and Adolescent Mental Health Services (CAMHs) were excluded from the data extraction & investigation, and therefore the first bin represents an age band of 18-25.

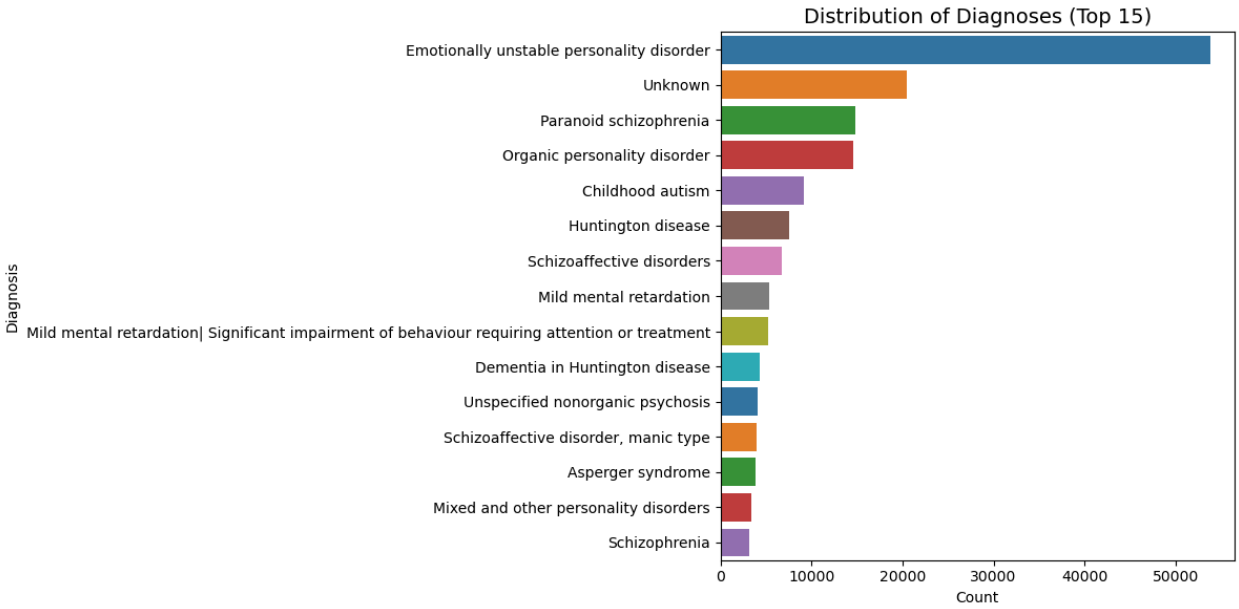



**Figure 5 - Distribution by Diagnosis**

*Figure 5* shows the distribution of Diagnoses found in the dataset. Emotionally Unstable Personality Disorder (EUPD) dominates the dataset, illustrating an imbalance. Although, some of the typical diagnoses observed in inpatient services are still represented with up to 20’000 entries. However, it is likely that this imbalance will have an effect on the expected contributory features for the prediction of violence, given the inherent behavioural differences expressed between diagnoses.

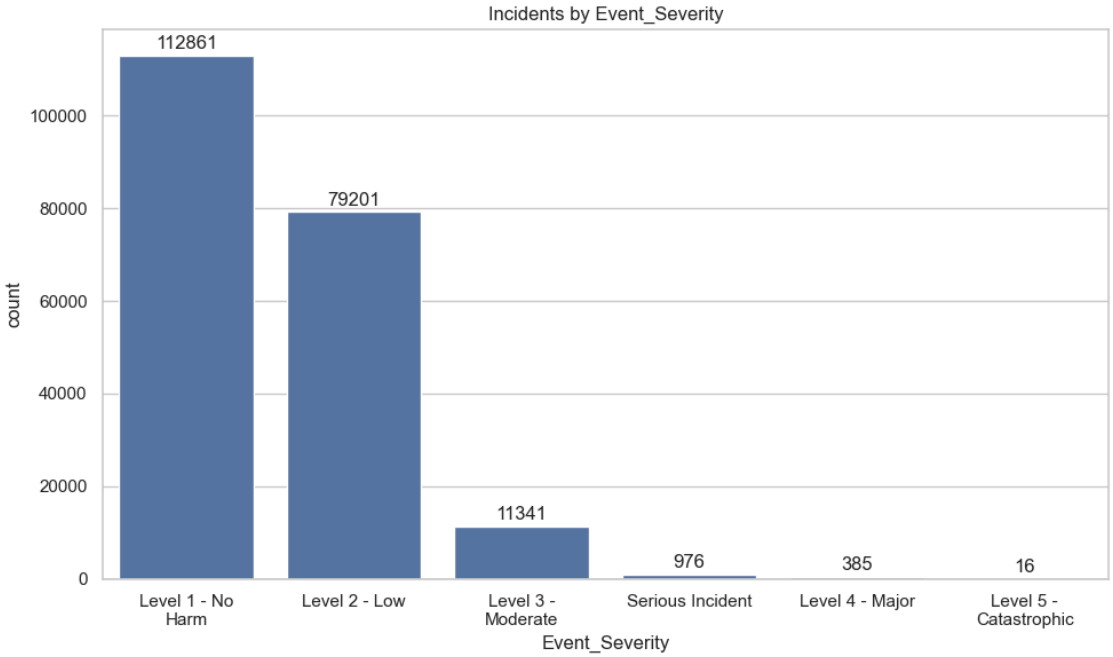



**Figure 6 - Distribution of Incidents by Severity**

*Figure 6* shows the distribution of incidents by severity in the dataset. Many mental health professionals utilise the Decision-Making Risk Matrix (Crisis Prevention Institute, 2021) in their day-to-day work. The matrix suggests that an event that has a high likelihood of happening but with no harm does not generally necessitate intervention. Whereas something that has catastrophic harm but has a low risk of happening still requires intervention. This provides context for the distribution of events in *figure 6.* Incidents that are at risk of serious harm are prepared for, and we see intervention long before they can occur. Given that the investigation intends to predict a violent incident, regardless of severity, the impact on model training should be minimal.

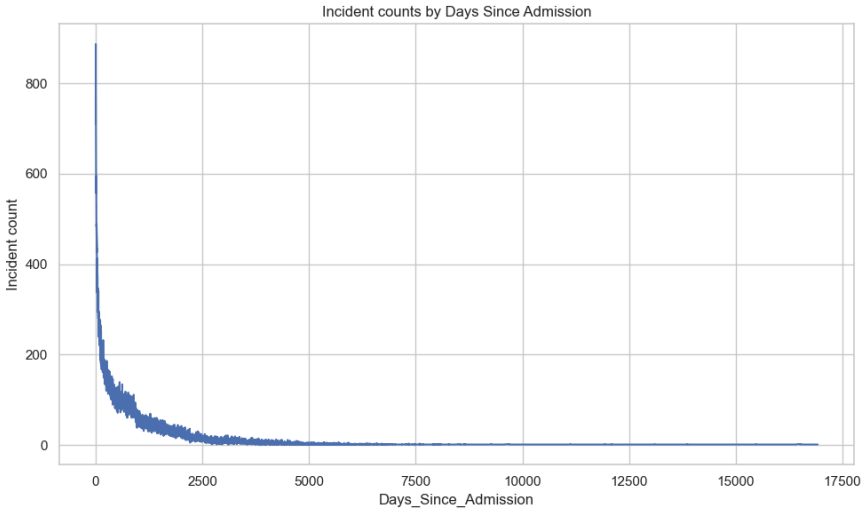



**Figure 7 - Incident Frequency over Admissions Visualisation**

*Figure 7* shows incidents against the days since a service user's admission. We can observe that almost all incidents are concentrated closer to the day of admission, with most on the first day. Thereafter, incidents tend to decrease as time progresses, likely due to acclimatisation to the locked ward environment as well as the commencement of recovery.

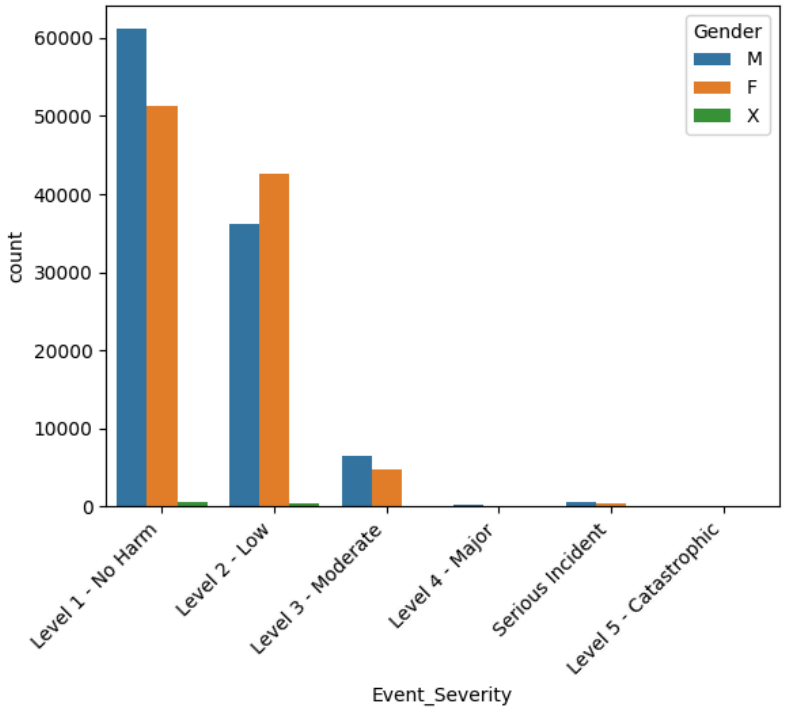



**Figure 8 - Distribution of Gender over Incident Severity**

The vast majority of incidents in *figure 8*, like *figure 6*, show the dominating proportion of incidents in the low to no-harm portion of the visualisation. However, splitting this by gender illustrates a slightly higher male share at the more serious levels. This could imply a necessity for ward-specific/demographic tailoring to ensure thresholds and responses are appropriate in a case of deployment. Overall, this distribution should provide a model focused on early-stage prevention, as the model will indirectly and inherently suggest intervention at stages where the risk presented is lower.

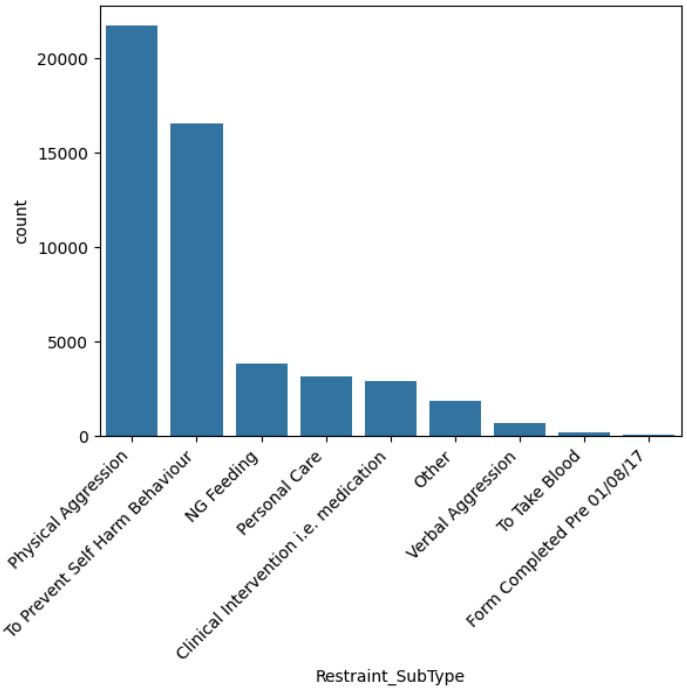



**Figure 9 - Distribution of Restraints & Justification**

*Figure 9* shows the distribution of the 'Restraint_SubType' feature, which functions as a descriptor for the purpose of a restraint. Physical Aggression is the most prevalent class in this feature, with self-harm as the second most common. This is in line with reality, given the most frequent purpose of restraint in an inpatient setting is to prevent harm due to aggression & self-harm (Shafti et al., 2023).

Established in the Seclusion Policy (St Andrews Healthcare, 2025), service users should not be placed in seclusion if their presentation is exclusive to self-harm behaviour. To increase the quantity of physical aggression values in the dataset, any restraint that led to a seclusion was flagged as a physically aggressive event. Although self-harm may have been present as well during this incident, some development will have led to expression of aggression, thus not introducing inaccuracy into this feature - provided decision-making was in line with the Seclusion Policy. There is consensus among professionals that, moreso in those suffering with EUPD, there is a common duality in the expression of these risk behaviours (Shafti et al., 2023).

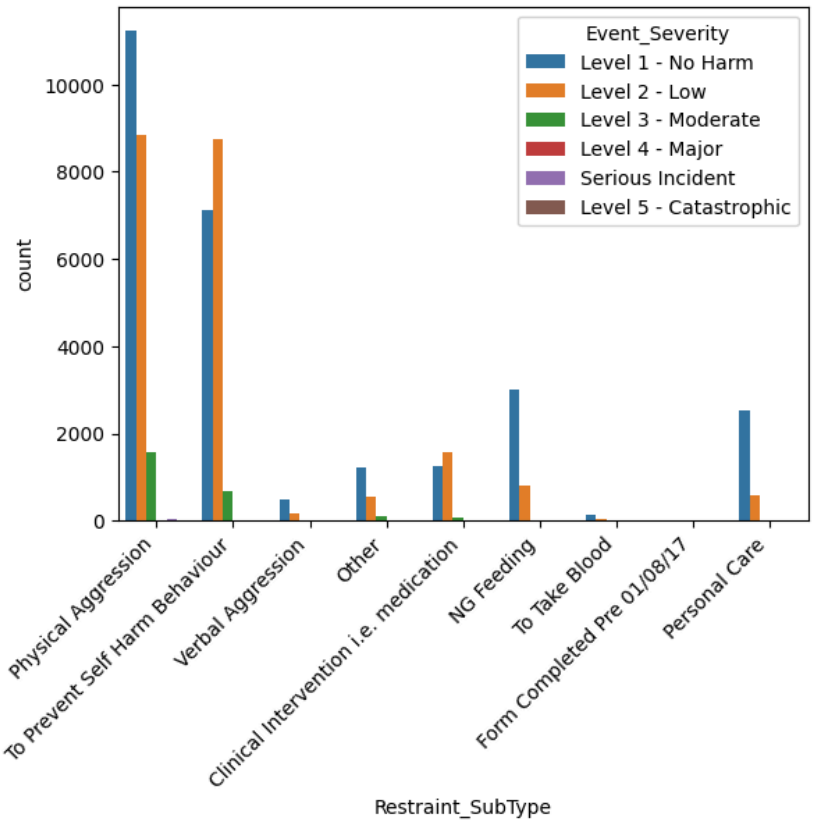



**Figure 10 - Distribution of Restraint Type by Event Severity**

*Figure 10* validates the suggestion that the occurrence of more serious incidents is observed predominantly in Physically Aggressive & Self-Harm type incidents.

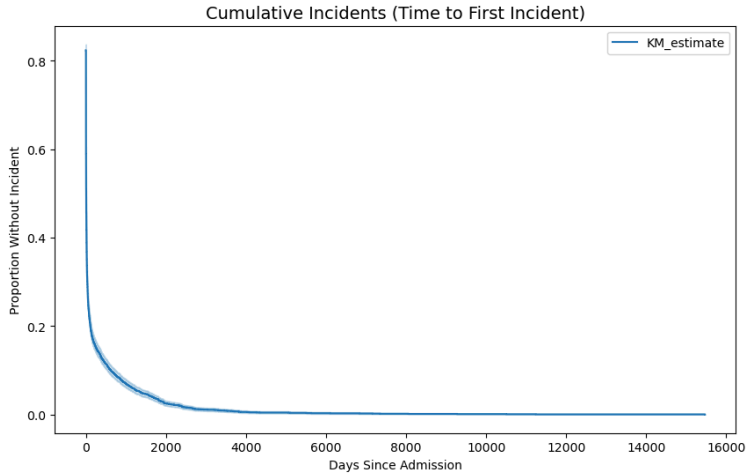



**Figure 11 - Time to First Incident Visualisation**

*Figure 11* shows a Kaplan-Meier 'survival' curve. The curve represents the proportion of those without incident since admission or the probability that an individual will go without incident over time; to 'survive' without incident. The curve suggests that there is a notable risk immediately post-admission, with the observable sharp decline at day 0. The relatively smooth curve into the tail shows the incorporation of those who effectively waited for a time before any risk behaviours were presented. The tail that tends towards zero suggests that long-term patients were unlikely to be incident-free in their journey through hospital. This is validated by the ethos that: should someone be in hospital for years and yet have no recorded incidents? Although this is not a bulletproof perspective when considering those with auxiliary care needs and vulnerabilities, the prevalence of more acute and episodic diagnoses in the dataset supports *figure 11*.

| Feature (Average) | Value (Hrs) |
|---|---|
| time_since_last_activity (hours) | 136.99 |

**Table 1 - Average Time since Last Activity**

We found that across the dataset, the average time between recorded meaningful activities is 137 hours or approximately 6 days. It is likely that this metric is right-skewed, given the proportion of patients that do not engage with offered activities at all. However, longer gaps between activities may suggest reduced engagement, ward acuity or some antecedents to an incident. Given that a patient may decline an activity, indicating a change in mood/mental state, or the activity may be cancelled by staff due to increased presented risk, produces a reverse causality. A feature relationship is established with observation_level to help mitigate this.

## 4.2 Model Performance Statistics

4.2.1 RNN Architecture

*The Hybridised Model*

The first model trialled in the investigation consisted of a SimpleRNN, LSTM & GRU layer each. The intention was to utilise the strengths of each architecture to reinforce a stronger performance.

During the progression of the investigation, the explicit performance metrics for this model were lost. However, we were able to calculate the F1 score and AUC from visualisations produced at that time, which are listed below.

| Metric | Value |
|---|---|
| AUC | 0.696 |
| F1 Score | 0.56 |

**Table 2 - Performance Metrics for Hybridised Model**

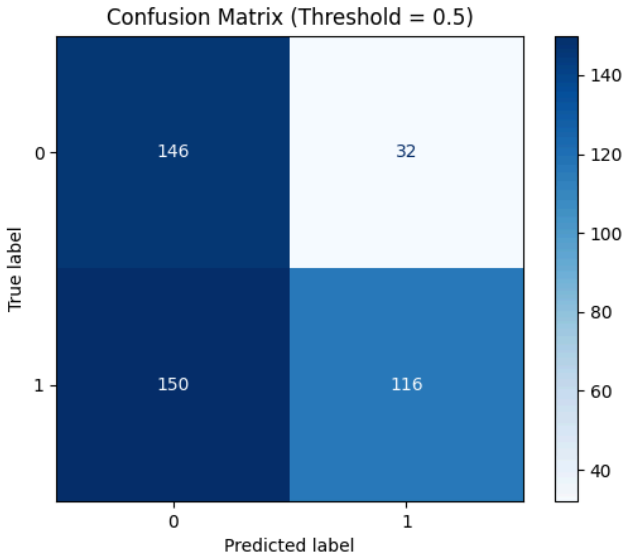



**Figure 12 - Confusion Matrix for Hybridised Model**

*Figure 11 & 12* show the results from the initial model trained. It achieved an AUC of 0.696 and an F1 Score of 0.56. *Figure 12* shows the model has strong specificity but also many false negative predictions, which would be unsuitable for purpose. It is likely that the inclusion of the SimpleRNN layer has hindered performance, given it's better suited for short-term dependencies and its inclusion likely introduced the vanishing gradient problem into the model (Pascanu et al., 2013), ultimately dismissing any long-range signals and diminishing the context available.

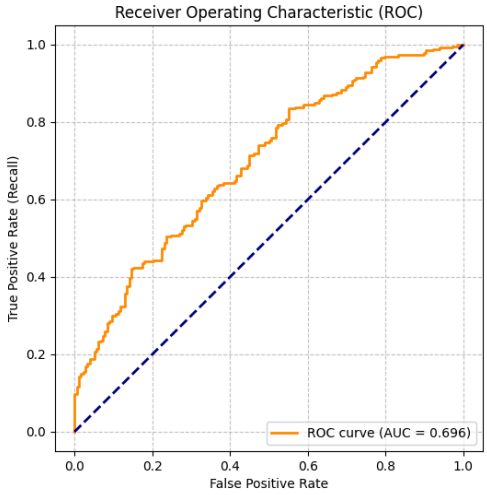



**Figure 13 - ROC AUC Curve for Hybridised Model**

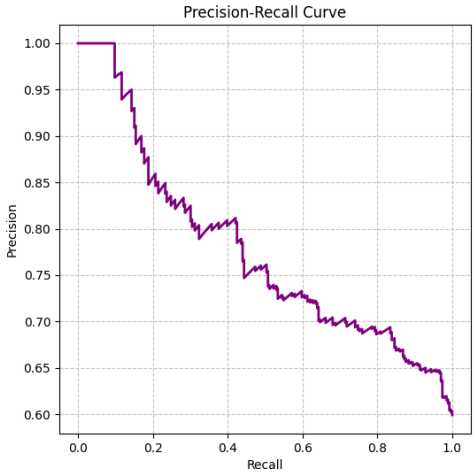



**Figure 14 - Precision-Recall Curve for Hybridised Model**

*Figure 13 & 14* show the ROC AUC and PR curves relating to the hybrid model. The diagonal seen in *figure 13* represents 0.5 AUC. We observe that the model slightly exceeds this, in line with the numeric AUC [0.696]. The model performs slightly better than chance in this aspect. The PR curve sits above the prevalence baseline, although only modestly. This suggests little practical utility in the context of this investigation.

*The Initial Homogeneous GRU Model*

This model utilised homogenous (sequencing) layers rather than the variation of layer types in the hybrid model before it, whilst also including Dropout layers and a Dense output layer.

| Metric | Value |
|---|---|
| **Brier Score** | 0.1 |
| **AUC** | 0.9064 |
| **F1 Score** | 0.85 |
| Precision | 0.9 |
| Recall | 0.81 |

**Table 3 - GRU Performance Metrics**

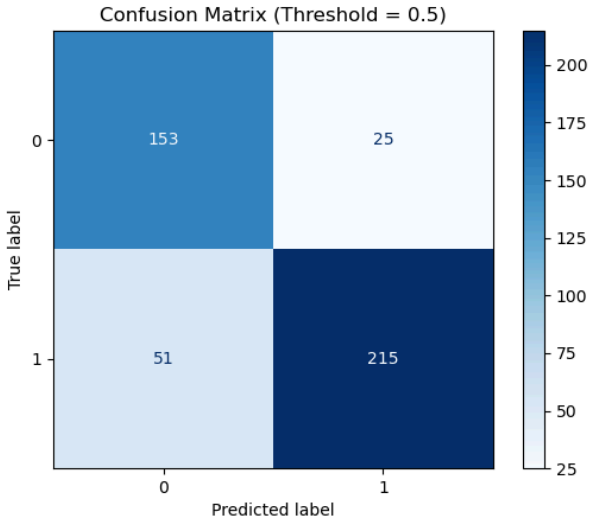



**Figure 15 - Confusion Matrix for GRU Model**

*Table 3 & Figure 15* show the performance metrics for the initial homogenous GRU model. The AUC score on the validation set of this model proved it had strong discriminatory power at 0.9064 and was able to distinguish effectively between violent and non-violent days. The F1 score achieved was 0.85, *figure 15* showing a slightly precision-leaning model. Thresholds were tested in quartiles, and 0.5 proved to be the most effective; however, it is likely that finer tuning could produce a better precision-recall balance. Compared to the previous model, the performance and viability in this case have improved significantly, with 1/10 predictions proving to be false alerts, statistically.

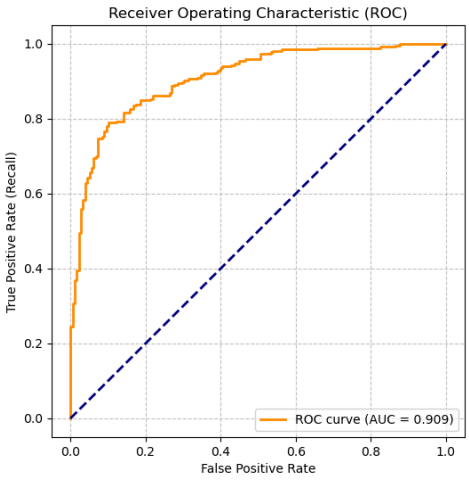



**Figure 16 - ROC Curve for GRU Model**

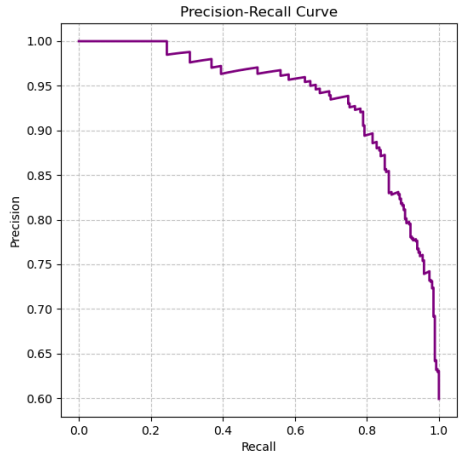



**Figure 17 - Precision Recall Curve for GRU Model**

*Figure 16* validates the AUC score from *Table 3,* illustrating strong discriminatory power. The curve suggests that GRU assigned slightly higher scores to true positives than negatives across thresholds. The Precision-Recall curve establishes itself well above the prevalence baseline for a majority of its range, although we see some deterioration in its mid-range falling as recall tends towards 1.0. This model appears to balance a high precision and recall well when considered in conjunction with the confusion matrix. Threshold tuning would be a viable next step to ensure the model's outputs are in line with a given clinician's risk management attitude, i.e. Recall or Precision leaning models.

*GRU Model with Attention Mechanism*

The final RNN presented is of similar structure to the previous GRU model but incorporates a manually defined attention layer to establish more appropriate weighting to timesteps before predicting.

| Metric | Value |
|---|---|
| **Brier Score** | 0.0884 |
| **AUC** | 0.9586 |
| **F1 Score** | 0.8974 |
| Precision | 0.9654 |
| Recall | 0.8383 |

**Table 4 - Performance Metrics for GRU w/ Attention Mechanism**

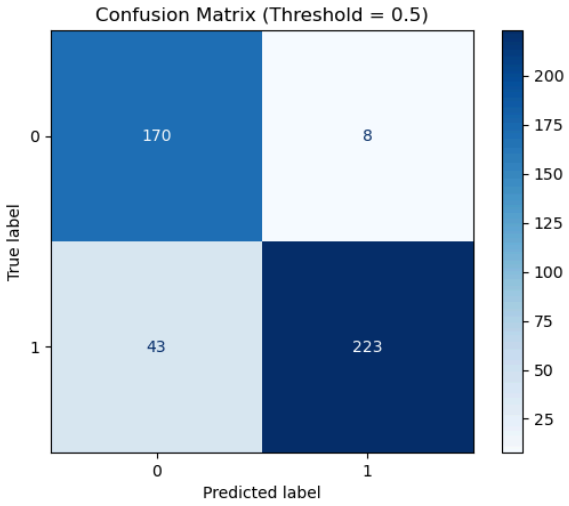



**Figure 18 - Confusion Matrix for GRU w/ Attention Mechanism**

*Table 4 & Figure 18* show the performance of the final model. The AUC demonstrated a very strong capacity to discriminate between classes. Further, the Brier score suggests a well-calibrated model when comparing probabilistic outputs to reality. The confusion matrix seen in *figure 18* demonstrates that only eight alerts were false alarms. Although the model is still precision leaning at this threshold, this can be adjusted to suit clinical preference easily. The metrics produced by this model suggest a clear advancement over all preceding models.

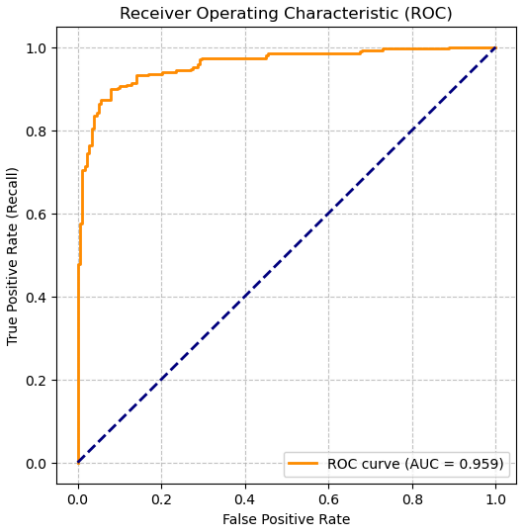



**Figure 19 - ROC Curve for GRU w/ Attention Mechanism**

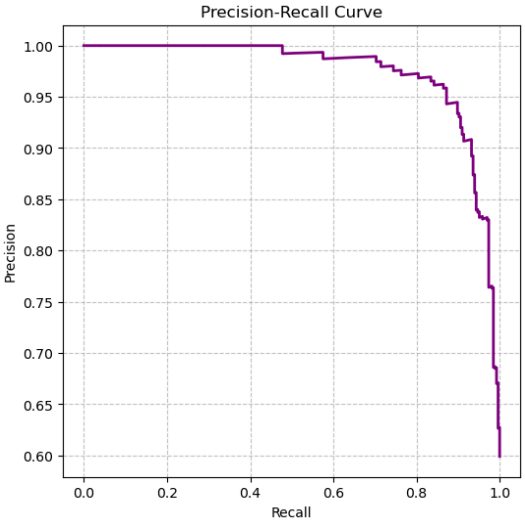



**Figure 20 - Precision Recall Curve for GRU w/ Attention Mechanism**

*Figures 19 & 20* show the AUC ROC curve and PR curve for the final GRU model. We can see that the ROC curve hugs the top left of the graph, further reinforcing its discriminatory power. There is a small amount of deterioration around the mid-range of the curve, not producing a true, perfect AUC curve. However, considering the AUC [0.959] and its graphical representation, this model has excelled in this aspect. The PR curve seen in *figure 20* sits far above the prevalence baseline. At the current threshold, 0.5, the model attains a precision value of 0.965 and a recall value of 0.838. For the purpose of an early-warning system, this is the most viable within the investigation, being the strongest model produced, with the possibility of threshold tuning to the suitability of the clinician(s).

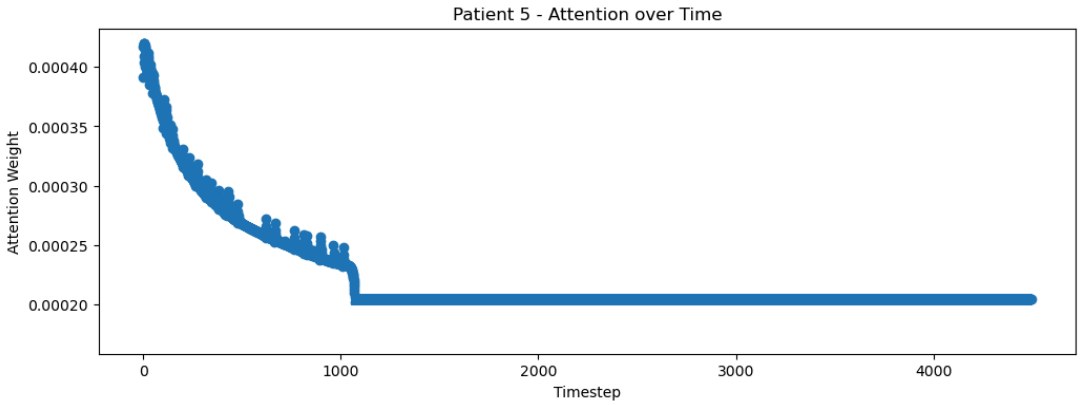



**Figure 21 - Example Attention Mechanism Visualisation**

*Figure 21* illustrates an example of the attention mechanism applying weights over a given patient's sequence. We can see the tail end of the curve is flat; this is likely due to the padded values filling the additional days. We can see some similarities in shape between the attention mechanism and the Kaplan Meier survival curve, *see figure 11*. This suggests that the attention mechanism has captured the relationship between timesteps in proximity to admission and the likelihood of risk behaviours being presented. The trend of the curve appears to have been gradually trending towards zero before the padded values began, further reinforcing the similarities between the figures. This may suggest that the attention mechanism aligns with reality in this case, assuming that *figure 11* is also representative of reality within the inpatient environment.

### 4.3 XAI Technique Outputs

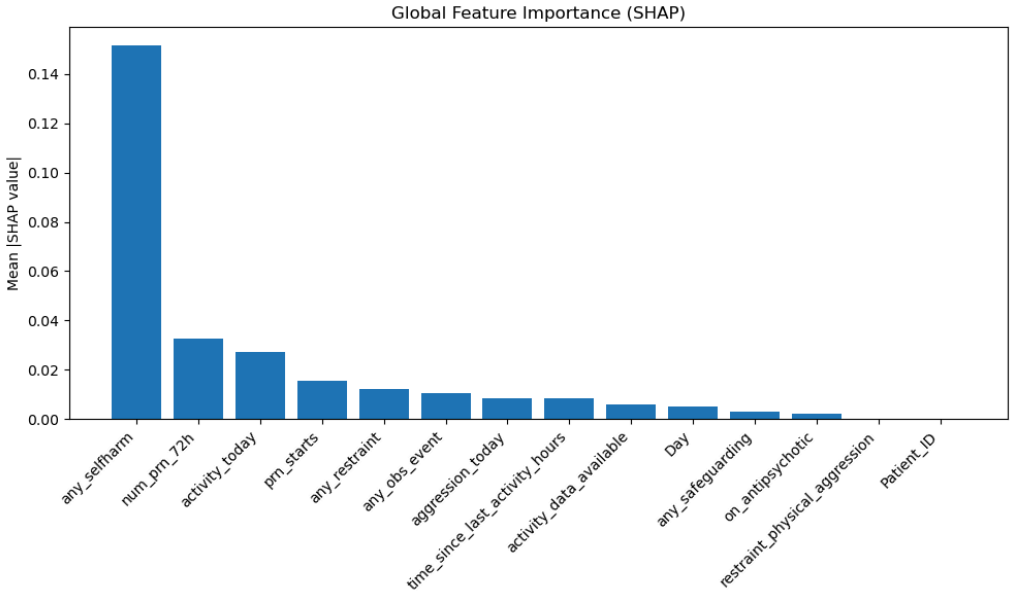



**Figure 22 - Global Feature Importance (SHAP)**

*Figure 22* shows the Global Feature Importance across the dataset when making predictions. On average, the predominant contributor was any incident of self-harm, followed by use of PRN medication in the last 72 hours and then whether they have engaged in any activities today. There is clarity regarding PRN and meaningful activities after discussion with mental health professionals, *see chapter 5.2*. Use of PRN often implies an unsettled mental state, and engaging in activities consistently should supposedly reduce some risk in presentation, as outcomes of these sessions are geared towards improving mood and mental state.

It is likely that the dominance of self-harm events as a contributor to predictions could be explained by *figure 5*. The predominant diagnosis in the dataset is 'Emotionally Unstable Personality Disorder'. Alongside the observed consensus amongst clinicians consulted with for this project, there is research that suggests a prevalence of self-harm behaviour in those suffering with Borderline Personality Disorder (BPD)/Emotionally Unstable Personality Disorder (EUPD) (Brickman et al., 2014). Further, there are studies suggesting there is a duality in the presentation of self-harm and violence and aggression amongst this demographic (Shafti et al., 2023), with the consulting clinicians for the project validating this sentiment. Given this dominance of EUPD in the dataset and the now verified duality of self-harm behaviours & violence and aggression, the model has established the contributory dominance of the self-harm feature in the same vein, aligning some of its predictions with the reality of this specific diagnosis.

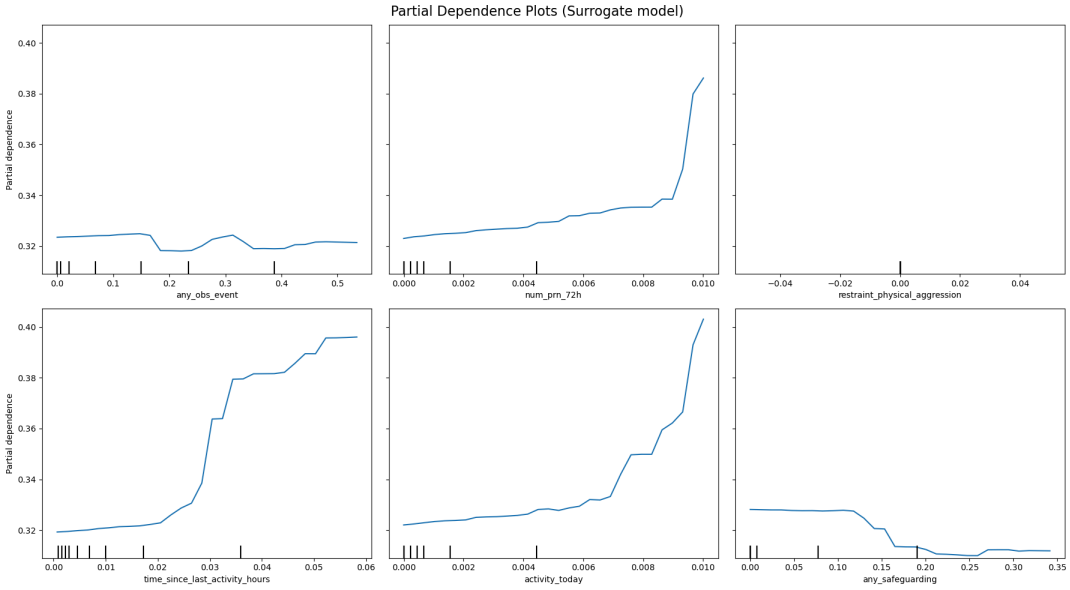



**Figure 23 - Partial Dependence Plots**

*Figure 23* shows partial dependence plots from a tree-based surrogate fitted to the final GRU model, indicating that recent medication activity and engagement gaps carry the strongest signal. Predicted risk increases with 'PRN within 72 hours' as well as with 'time since last activity'. Whereas observation events contribute little where the other features are consistent. PDP's assume partial independence; they are typically used in tandem with SHAP outputs.

## 4.4 UI Widget

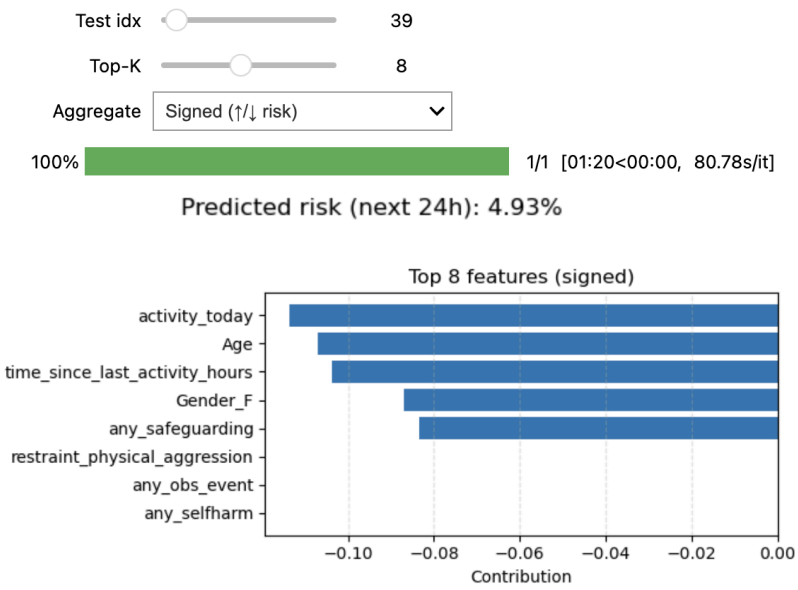



**Figure 24 - Model Widget Output #1**

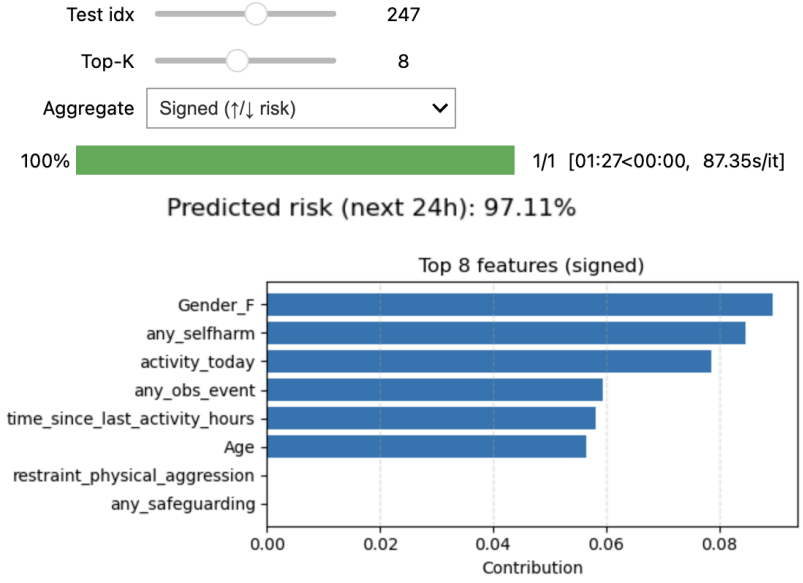



**Figure 25 - Model Widget Output #2**

*Figures 24 & 25* show a primitive per-patient prediction output UI with explanations, with *figure 24* predicting 'Patient 39' and *figure 25* predicting 'Patient 247'. *Figure 24* shows the predicted risk for the relevant patient [4.93%] and what contributed to this prediction. We observe that some static demographics are contributing; however, the dynamic activity-related features are strong contributors as well. *Figure 25* shows that some of the top contributors are self-harm, recency of activity & change in observation level and are predicted at a ~97% risk of aggression at this time.

The magnitude in which the features are represented illustrates how they've contributed with negative contribution values producing lower risk probabilities and positive contribution values increasing them. The figures also illustrate that, despite self-harm being the dominant global contributor, it is not necessarily present in every prediction. This suggests some capacity to differentiate in this case, and not a dominating 'master feature' that decides the probability outputs universally.

### 4.5 Transformer Tinkering

| Metric | Value |
|---|---|
| Brier Score | 0.0551 |
| AUC | 0.8228 |
| F1 Score @ 0.175 Threshold | 0.3535 |

**Table 5 - Transformer Performance**

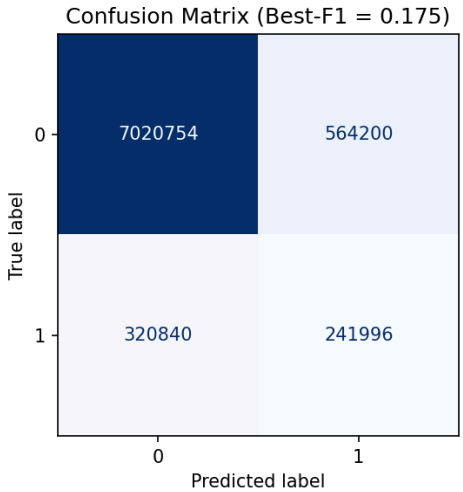



**Figure 26 - Confusion Matrix for Transformer**

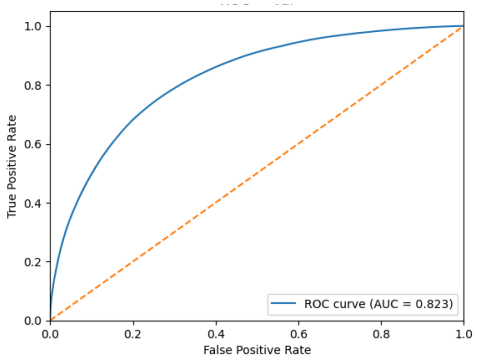



**Figure 27 - ROC AUC Curve for Transformer**

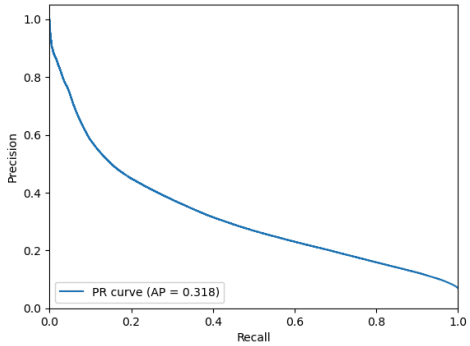



**Figure 28 - Precision Recall Curve for Transformer**

*Table 5, Figures 26, 27 & 28* each illustrate the performance of the Transformer model.
At its best operating threshold, the transformer achieves an AUC of 0.823, F1 score of 0.354 and Brier score of 0.055, established in *table 5*. Most alerts made are false, as over half of the true positive days are missed, as seen in the confusion matrix. The AUC suggests a moderate ability to discriminate, whereas the PR curve drops very quickly. The model's precision is high, but it performs poorly with respect to recall; this is a model that leans heavily to the left of our confusion matrix. It would seem that balancing for this model was insufficient, as the model appears to have found it safer to predict ‘no incident’, as that is what was more represented in the dataset overall.

This model in this state is observably inferior to the model produced in the main workflow of the investigation. The simpler binary representation utilised may have contributed to its lack of performance, as well as the 90-day forecasting target. However, the main reason the Transformer model performed poorly is due to the time constraints of the investigation and the lack of computational resources to mitigate this obstacle.

# 5 Discussion

## 5.1 Machine Learning Performance Discussion

### 5.1.1 Model Comparison

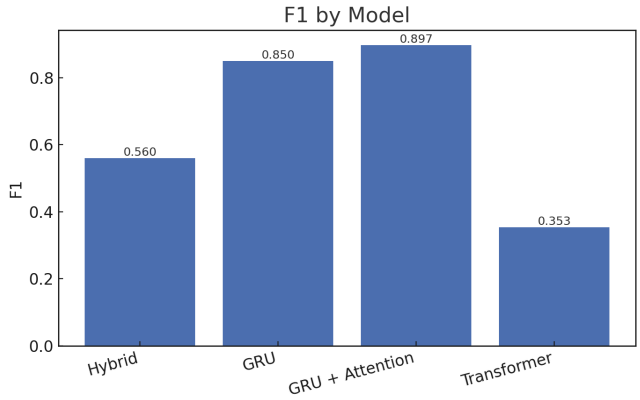



**Figure 29 - Model Comparison (F1 score)**

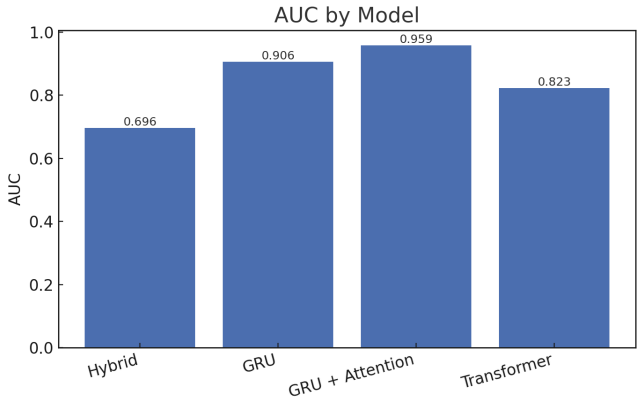



**Figure 30 - Model Comparison (AUC)**

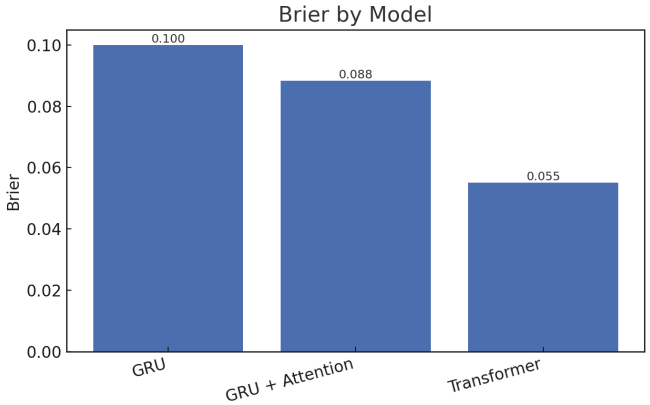



**Figure 31 - Model Comparison (Brier)**

*Figures 29, 30 & 31* compare the performance between each type of model testing in the investigation.

Contrasted with the beginning of training, performance improved greatly. Following the overall underperformance of the hybrid model, we moved to a homogeneous GRU model before settling on a similar model with an additional attention layer. The initial hybrid model offered a moderate separability, only slightly more than flipping a coin, overall proving to be an unsuitable predictor for risk. The homogeneous model performed drastically better, demonstrating strong discriminatory power and a relatively balanced F1 score by precision and recall. At the conclusion of model training, the GRU model outperformed all that came before it, with the highest Precision, Recall, F1 score, AUC and Brier amongst all predictors. These results suggest that gating and attention to temporally important events are crucial to prediction of this type. Further, validating the sentiment that recency of events will have more impact due to a more present illustration of mood and mental state.

### 5.1.2 Thresholds & Operating Points

Given the intended use of machine learning in this context would be for an early-warning predictor for risk, the threshold choice matters as much as ranking. The GRU-A model captures most positives correctly with only four false positives, whereas we observe fourty-three false negatives in the final model at a 0.5 threshold. The Precision-Recall curve suggests that there is room to trade off between Precision and Recall in this case to slightly alter these values. Importantly, this can be done in conjunction with clinical staff to tailor alerts to their desired tolerance.

### 5.1.3 Calibration

Brier scores highlight that the probability quality is usable for decision support: the GRU-A model's Brier score at 0.088 suggests well-calibrated risk estimates. This is essential if the probability output in its raw form is to be used meaningfully. Notably, the Transformer achieved an even lower Brier score of 0.055 but a remarkably poorer F1 at 0.35.

### 5.1.4 Contributions (XAI)

Global SHAP and PDP analyses converge on dynamic and recent signals found in the dataset. Self-harm events, PRN use, and time since last activity prove influential. PRN consistently increases risk estimates as well as longer gaps since activity - 'activity_today' values in the dataset may be due to a potential reverse causality, so it is considered as associative.

### 5.1.5 Limitations and Generalisability

The data utilised in the investigation has been supplied by a single provider and likely reflect local demographics, so external validity requires caution. Days where no activity or incident took place were imputed to create a consistent timeline can be fed into the models. This introduces some risk of inaccuracy as the values are not as reliable as timesteps about those days from clinical staff. Also, at the conclusion of training, it became clear that the original dataset labels were majorly intact, which would damage the interpretability of explainability techniques for anyone not familiar with the investigation and this report.

The skew towards EUPD diagnosis in the dataset also suggests a poorer predictive quality for some of the other underrepresented service users in the dataset. Although this fact enlightens regarding the strong contribution of self-harm events to predictions across the dataset, it also suggests that models tailored for specific ward environments or acuities may be a more suitable progression for predicting risk in inpatients rather than a predictor for all.

### 5.1.6 Transformers

Transformers, in principle, as an architecture, could prove to be more effective than the primary models produced in this investigation. Despite its poor classification performance in this case, the inherent self-attention that transformers have can model long-range, non-local dependencies, e.g. clustering of incidents around changes in medication. Transformers don't suffer from the vanishing gradient problem and perform well over long sequences. Had this investigation not been limited by time and had the appropriate computational resources to mitigate the time complexity of training within the Transformer architecture, there is a high chance that performance outputs would have been greater.

Should investigation continue on temporal architectures to predict risk, akin to this investigation, it is likely that Transformers will outperform the Recurrent Neural Network Architecture overall.

## 5.2 Clinical Discussion

5.2.1 Clinical Insights

To ensure that the investigation was clinically meaningful, we endeavoured to consult with clinicians in the sector across multiple disciplines to garner input and insight on the project at different intervals of its progression. The ensemble of professionals includes Occupational Therapists, Qualified & Experienced Nursing and Managerial staff as well as Consultant Psychiatrists. Individual summaries were created for each individual.

*Relevance and value*; all those spoken with agreed that predictive tools in general are prudent and necessary in the process of managing violence and aggression. Both veteran RMN's highlighted the shortcomings of static risk prediction tools, like the HCR-20, when considering the dynamic and unpredictable nature of aggressive incidents in an inpatient environment. Both psychiatrists commented on the few existing, validated short-term prediction tools. Finally, occupational therapy highlighted the practical benefits with regards to resource allocation and proactive care from an early warning system that utilised the outputs from this investigation.

*Prediction window*: There was a general consensus among clinicians that a 24-hour window is meaningful and appropriate for predicting violent risk. There was further discussion regarding the potential for shorter windows; however, it was agreed that anything shorter than 12 hours would prove difficult. This is due to the nature of the data collection that is necessary for the model to work; Datix incidents and meaningful activities may not be completed until the end of a given shift.

*Existing prediction tools*: many of the clinicians reported tools being tick-box exercises. Further, there was prevailing discussion about how they can be overly subjective, some stating that the human factor contributes undeniable bias in the process of their completion.

*Interpretability of outputs*: Opinion on whether a traffic light system, raw probability output or scoring out of ten was divided amongst clinicians. OT felt traffic lights are simple and interpretable for frontline staff. Although the discussion was unable to come to an appropriate solution for where to set thresholds for each of the three traffic light boundaries. Overall, the prevailing preference was for raw probability scores. The following discussion in support of this was that scores would promote clinical judgement and begin to relate a given percentage to the individual. Effectively, scaling the outputs proposed by the model to an individual's behaviour further enforces personalised use of the tool per patient. Every clinician agreed that some kind of narrative explanations for predictions would be necessary to ensure the tool is not biased and further promote its credibility.

*Risks & ethical concerns*: Over-reliance and misunderstanding of design intention were a prevailing concern. There was extensive discussion regarding how staff may avoid service users with high risk scores, whereas the development of a tool in line with the findings of this investigation would be to encourage proactive care and intervene before risk behaviours are presented. Further, one psychiatrist stated that uninvestigated systemic variables may promote bias inherently, i.e. a given demographic may not engage with meaningful activities. Since we've identified that the model utilises the recency of meaningful activities in its predictions, this may result in higher risk probability outputs. Time was spent attempting to balance Precision and Recall in a final model. Most felt that a higher recall was preferable when considering risk management, the sentiment being that false alarms are better than no alarms. However, one RMN stated that a focus on precision would discourage over-reliance and promote clinical judgement alongside potential use of this tool. All agreed that balancing of these metrics is important to establishing trust and credibility amongst staff, such as a predictor of this type.

*Explainability & interpretability*: We observed universal agreement on the necessity of explainability for predictions. Clinicians must be aware of what has influenced predictions to create synergy with their own clinical judgement and further enable communication with service users regarding the risk they may be presenting. Some clinicians appeared dubious about whether narrative explanations would complicate use. However, the dominating perspective was an advocacy for explicit interpretability when considering clinical (and ethical) acceptance. Further, the former quality assessor confirmed that explainability would be mandatory for consideration by the CQC - but that they would likely support a quantitative, evidence-based predictor.

*Considering potential future implementation*: All agreed that additional admin tasks to make use of a tool powered by machine learning would be a major drawback. Many believed that supporting training alongside implementation would help mitigate misuse. Two acknowledged that the way to truly evaluate and validate the performance of tools would be for a ward to pilot them.

Across all interviews, there was an apparent strong support for the principle and implementation of machine learning tools for prediction of risk, should it be developed to the standard and expectations previously established for any tool utilised in healthcare. Comment was made consistently regarding the drawbacks of existing risk assessment tools, such as subjectivity and staticism. Enthusiasm was apparent amongst all clinicians, but not without concern for misuse and poor implementation. Overall, the common perspective among all had an emphasis on balanced implementation, embedding into existing systems and processes,

ensuring the facilitation of training and governance, but most importantly, maintaining clinical judgement as central to any risk management.

5.2.2 Clinical Implications

The introduction of machine learning tools in this sector for prediction of violent incidents carries notable implications. Many clinicians recognise the potential for improved, proactive care and safety. An effective model operating on a 24-hour window could support timely de-escalation, targeted engagement and has the potential to contribute to resource allocation.

This aligns with some perspectives regarding existing risk assessment tools, such as Brøset and HCR-20 and their static and subjective [to the assessor] nature. Further, we saw comment on the drawbacks of how a static assessment will struggle in some aspects when assessing something as dynamic as risk in an inpatient setting. A data-driven model that considers multiple variables, such as Incident History, Recent PRN & Meaningful Activities, *could* provide a responsive and objective supplement to clinical judgement.

There was special attention given to discussing how it would be used in practice. A recurring theme was of over-reliance: that clinicians may treat predictions as definitive, withdraw therapeutic contact or potentially use it as sole justification for restrictive measures/interventions. Should there not be an appropriate mindset established in staff upon adoption of a tool such as this, there is risk of disengagement from staff, which can also trigger a crisis event. There is potential for the tools that are developed without an awareness of bias to entrench systemic inequalities rather than relieve them.

Clinical staff would need training on interpreting probabilities, even narrative explanations and how to effectively and appropriately use this information. Fundamental explanation of model design and design intention, communicated in an easily understandable manner, would support understanding of the tool and further build credibility. Overall transparency would be a must for assisting clinicians when incorporating outputs into the formulation of their clinical judgement, as well as communicating clearly with patients about what has led to any decision-making. Further, transparency assists in the demonstration of evidence-based predictions for regulators such as the CQC.

Governance and awareness of culture is crucial. Embedding of tools into existing data collection pipelines is necessary. No one demonstrated a positive response when considering being faced with more admin tasks. A reliable model could be supportive in multidisciplinary team (MDT) reviews and could encourage attention to be placed on a given individual with unnoticed, escalating risk. Transparency about the tool's limitations will also promote more effective use of a tool if the relevant staff are aware where a model falls short when considering its outputs, as well as manage expectations and maintain credibility.

The prevailing concern is not regarding the implementation of an effective model but rather the implementation without adequate safeguards. If integrated with governance, explainability and training, then the tool could represent an innovative step forward in patient and staff safety. Not only could an effective model reduce incidents, but it also has the potential to reinforce and evidence the protective effects of meaningful activities and therapeutic engagement. Overall, prompting a shift from the recent & prevailing reactive approach to risk management to a proactive one.

# 6 Conclusion

The project set out to investigate whether routinely collected electronic health records, including Datix incidents, could support an early warning, temporal, machine learning model. The findings supported this in principle. A GRU-based model with a manually defined attention mechanism achieved proficient discrimination, useful calibration and high precision & recall [AUC: 0.959, Brier: 0.1, F1: 0.897]. The earliest temporal model tested performed moderately due to shortfalls in architecture relating to the length of admission sequences. The Transformer, which was created for exploratory purposes and to add depth to the research, demonstrated feasibility but underperformed in contrast to the later RNNs produced. Overall, these findings showcase that temporal deep learning models that make use of dynamic, recent features and their variance over time can provide informative 24-hour risk estimates when paired with appropriate thresholding and calibration. Although this is only verified statistically. Following on from the models produced, the next step would be to verify and tune a model via piloting and then contrasting predictions with the manifestation of risk observed in reality by service users.

Considering the chronology of the investigation reflects some shortfalls. The dataset has a prevalence of EUPD diagnoses within, although other diagnoses are not absent; it is likely predictions for the minority demographics will not be as reliable. This may suggest a more effective model that reduces scope to specific ward environments and acuities. Further, there is a dominance of the ‘White British’ ethnicity in the data, likely due to the location in which the majority of the data was collected; this may also reflect in poorer quality predictions for minorities, despite explainability statistics not relying on ethnicity so heavily. The use of explainability for machine learning remains crucial, as validated by clinicians and the investigation alike. Discussions around implementation stand firm, regarding the use of machine learning in conjunction with clinical judgement and supported with the appropriate governance. External validation alongside the aforementioned prospective piloting of predictive tools are the logical next steps for an investigation of this type.
The investigation appears to have contributed, although minimally, to the foundation of proactive, machine learning powered, early warning systems in the mental health sector. Further, the potential for anticipatory outputs to reinforce early & individualised responses to escalating service users also aligns with the ethos set out by NICE in *NG10* and *QS154, see 2.3.2*. The benefits of a morally & holistically developed, governed and tested tool in the future could undeniably support a less-restrictive and safer environment for service users, staff members and visitors within a secure mental health setting.

# 7 Acknowledgements

Firstly, I would like to acknowledge the difficulties this investigation and I have faced over its chronology. Near the commencement of the project, I lost my grandfather, and this affected me greatly over the coming months. This coincided with the first application to research deadline with St Andrews, which was missed. The following month, the application was submitted and accepted. Thereafter, I awaited data extraction to begin the research. Through no fault of anyone involved in or adjacent to the project, the data was released near the beginning of September with the provisional UoP deadline of mid-late September.

It is at this point I would like to acknowledge, appreciate and apologise to the people closest to me, for I have endeavoured to complete the project to an appropriate standard in ~3 weeks, and I have likely not been a pleasure to be around during this time. However, if you are reading this, it means: I did it.

I would like to express my gratitude to my supervisor at the University of Portsmouth for their guidance, encouragement, and feedback throughout this project. Their expertise and advice have been important in shaping the direction of this work.

I am also profoundly grateful to St Andrew's Healthcare for their collaboration and support in providing access to the data that made this research possible. Special thanks to the clinical experts who shared their insights during conversations regarding project progress.

*cont.*

I would particularly like to acknowledge the following individuals for their contributions and support:

- **Dr Rinat Khusainov** - [Academic Project Supervisor; continued advice, support and assistance in getting the project underway despite hurdles and difficulties]
- **Dr Ash Roychowdhury** - [Consultant Psychiatrist,  Relevant Clinician & Lead Innovator; for the meaningful conversations about the implications of ML predictors & implications and risks of implementation]
- **Dr Samuel Amo-Koryanke** - [Consultant Psychiatrist; clinical expertise and discussion participation]
- **Annym Adams, *Msc*** - [Lead Occupational Therapist; input on patient activity data and interpretation as well as overall informed clinical insights]
- **Simla Aldridge, *Bsc (Hons)*** - [Veteran RMN, CBT Therapist, Advanced Clinical Lead Skills Facilitator (for Self-Harm & Suicide); guidance on staff policy, training, NICE guidance and de-escalation practice]
- **Ross Callaghan, *Bsc (Hons)*** - [RMN, PICU Ward Manager; for extensive insight into staff response and dynamics]
- **Dean Aldridge, *Bsc (Hons)*** - [Veteran RMN, Deputy Ward Manager, Internal Quality Assessor, Commissioner; clinical insights and risk aware perspectives, positive outlooks on tool implementation]

I would like to acknowledge the contributions of these Consultant Psychiatrists, Occupational Therapists, and Mental Health Nursing Professionals who participated in discussions and provided perspectives that grounded this research in real-world clinical practice. Their input has been vital in ensuring that, even though this is an academic exercise, the study retained relevance regarding the needs and realities of mental health services.

I especially would like to acknowledge all those whose prior work in the fields of machine learning, psychiatry, and mental health safety informed and inspired this research.

Specifically, I would like to extend special thanks to my good friend Albert. His knowledge and consultation on machine learning methods truly kept me grounded in the flurry of research and model training that has been this project.

Lastly, I would like to thank again, my partner, family and friends for their patience, encouragement, and support throughout the project.

And, thank you Grandad Dave, for turning up in the end before you went away.

With support & collaboration from






# 8 References

Almvik, R., Woods, P., & Rasmussen, K. (2000). The Brøset Violence Checklist. *Journal of Interpersonal Violence*, *15*(12), 1284–1296. https://doi.org/10.1177/088626000015012003

Appleby, L., & NCISH. (2021). *National Confidential Inquiry into Suicide and Safety in Mental Health*. https://www.hqip.org.uk/wp-content/uploads/2021/05/NCISH-Annual-Report-2021.pdf

Appleby, L., & NCISH. (2025, February 13). *Suicide and safety in mental health (NCISH) – HQIP*. HQIP. https://www.hqip.org.uk/resource/mental-health-ncish-feb25/

Blitz, R., Storck, M., Baune, B. T., Dugas, M., & Opel, N. (2021). Design and Implementation of an Informatics Infrastructure for Standardized Data Acquisition, Transfer, Storage, and Export in Psychiatric Clinical Routine: Feasibility Study. *JMIR Mental Health*, *8*(6), e26681. https://doi.org/10.2196/26681

Brickman, L. J., Ammerman, B. A., Look, A. E., Berman, M. E., & McCloskey, M. S. (2014). The relationship between non-suicidal self-injury and borderline personality disorder symptoms in a college sample. *Borderline Personality Disorder and Emotion Dysregulation*, *1*(1), 14. https://doi.org/10.1186/2051-6673-1-14

Brownlee, J. (2020, August 25). *Use Early Stopping to Halt the Training of Neural Networks*. Machine Learning Mastery. https://machinelearningmastery.com/how-to-stop-training-deep-neural-networks-at-the-right-time-using-early-stopping/

Charity Commission for England and Wales. (2023). *Compliance toolkit: protecting charities from harm Chapter 2: Due diligence, monitoring and verifying the end use of charitable funds Tool 4: Risk management Risk matrix*. https://assets.publishing.service.gov.uk/media/5a8008a4e5274a2e8ab4dcad/Tool_4.pdf

Choi, E., Taha Bahadori, M., Kulas, J., Schuetz, A., Stewart, W., & Sun, J. (2018). *RETAIN: An Interpretable Predictive Model for Healthcare using Reverse Time Attention Mechanism*.

https://proceedings.neurips.cc/paper_files/paper/2016/file/231141b34c82aa95e48810a9d1b33a79-Paper.pdf

Chung, J., Gulcehre, C., Cho, K., & Bengio, Y. (2014). *Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling*. ArXiv.org. https://arxiv.org/abs/1412.3555

Crisis Prevention Institute. (2021). *The Decision-Making Matrix : The Tool That Helps Reduce Behavioral Risk Factors*. Www.crisisprevention.com. https://www.crisisprevention.com/blog/behavioral-health/risk-behaviors/

Daffern, M., & Ogloff, J. (2015). *The Dynamic Appraisal of Situational Aggression (DASA)*. Www.swinburne.edu.au. https://www.swinburne.edu.au/research/centres-groups-clinics/centre-for-forensic-behavioural-science/our-publications-products/dasa/

DOUGLAS, K. S., & WEBSTER, C. D. (1999). The HCR-20 Violence Risk Assessment Scheme. *Criminal Justice and Behavior*, *26*(1), 3–19. https://doi.org/10.1177/0093854899026001001

Flach, P., & Kull, M. (2015). *Precision-Recall Curves: Analysis Done Right*. Neural Information Processing Systems; Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2015/hash/33e8075e9970de0cfea955afd4644bb2-Abstract.html

Garriga, R., Mas, J., Abraha, S., Nolan, J., Harrison, O., Tadros, G., & Matic, A. (2022). Machine learning model to predict mental health crises from electronic health records. *Nature Medicine*, *28*(6), 1240–1248. https://doi.org/10.1038/s41591-022-01811-5

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. Www.deeplearningbook.org. https://www.deeplearningbook.org

Hassija, V., Chamola, V., Mahapatra, A., Singal, A., Goel, D., Huang, K., Scardapane, S., Spinelli, I., Mahmud, M., & Hussain, A. (2023). Interpreting Black-Box Models: a Review on Explainable Artificial Intelligence. *Cognitive Computation*, *16*(1). https://doi.org/10.1007/s12559-023-10179-8

Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, *9*(1), 1–42. https://doi.org/10.1162/neco.1997.9.1.1

James, M., Painter, J., Buckingham, B., & Stewart, M. W. (2018). A review and update of the Health of the Nation Outcome Scales (HoNOS). *BJPsych Bulletin*, *42*(2), 63–68. https://doi.org/10.1192/bjb.2017.17

Keras. (2023). *Working with RNNs | TensorFlow Core*. TensorFlow. https://www.tensorflow.org/guide/keras/working_with_rnns

Kingma, D. P., & Ba, J. (2014, December 22). *Adam: A Method for Stochastic Optimization*. ArXiv.org. https://arxiv.org/abs/1412.6980

Lim, B., Arik, Sercan O, Loeff, N., & Pfister, T. (2019). *Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting*. ArXiv.org. https://arxiv.org/abs/1912.09363

Linaker, O. M., & Busch-Iversen, H. (1995). Predictors of imminent violence in psychiatric inpatients. *Acta Psychiatrica Scandinavica*, *92*(4), 250–254. https://doi.org/10.1111/j.1600-0447.1995.tb09578.x

Malato, G. (2021, May 19). *Hyperparameter tuning - Grid search and Random search*. Your Data Teacher. https://www.yourdatateacher.com/2021/05/19/hyperparameter-tuning-grid-search-and-random-search/

Murphy, A. (1973). *A New Vector Partition of the Probability Score*. American Meteorological Society. https://doi.org/10.1175/1520-0450(1973)012%3C0595:ANVPOT%3E2.0.CO;2. Brier Score.

National Institute for Health and Care Excellence. (2015, May 28). *Violence and aggression: short-term management in mental health, health and community settings | Guidance | NICE*. Nice.org.uk; NICE. https://www.nice.org.uk/guidance/ng10

National Institute for Health and Care Excellence. (2020). *Reducing the risk of violent and aggressive behaviours | quick guides | social care | NICE communities | about*. NICE. https://www.nice.org.uk/about/nice-communities/social-care/quick-guides/reducing-the-risk-violent-and-aggressive-behaviours

National Institute for Health and Care Excellence. (2022, September 7). *Overview | Self-harm: Assessment, Management and Preventing Recurrence | Guidance | NICE*. Www.nice.org.uk; National Institute for Health and Care Excellence. https://www.nice.org.uk/guidance/NG225

NHS England. (2015). *HCR-20 v3 Assessing Risk for Violence*. https://madeinheene.hee.nhs.uk/Portals/19/1%20HCR%2020%20V3%20Introduction-%20Dr%20Barlow%20_1.pdf

NICE. (2017, June 29). *Overview | Violent and aggressive behaviours in people with mental health problems | Quality standards | NICE*. Nice.org.uk; NICE. https://www.nice.org.uk/guidance/qs154

Ogloff, J., & Daffern, M. (2015). *Practice Development Improving Patient and Staff Safety DYNAMIC APPRAISAL OF SITUATIONAL AGGRESSION- INPATIENT VERSION*. https://www.rcpsych.ac.uk/docs/default-source/improving-care/nccmh/reducing-restrictive-practice/resources/dasa-information.pdf?sfvrsn=d779798b_2

Palacio, S., Lucieri, A., Munir, M., Hees, J., Ahmed, S., & Dengel, A. (2021). *XAI Handbook: Towards a Unified Framework for Explainable AI*. ArXiv.org. https://arxiv.org/abs/2105.06677

Palmstierna, T., & Wistedt, B. (1987). Staff observation aggression scale, SOAS: Presentation and evaluation. *Acta Psychiatrica Scandinavica*, *76*(6), 657–663. https://doi.org/10.1111/j.1600-0447.1987.tb02936.x

Pascanu, R., Mikolov, T., & Bengio, Y. (2013). On the difficulty of training Recurrent Neural Networks. *ArXiv:1211.5063 [Cs]*. https://arxiv.org/abs/1211.5063

R Almvik, & Woods, P. (1999). Predicting inpatient violence using the Brøset Violence Checklist (BVC). *The International Journal of Psychiatric Nursing Research*, *4*(3), 498–505.

https://www.researchgate.net/publication/12849163_Predicting_inpatient_violence_using_the_Broset_Violence_Checklist_BVC

Saito, T., & Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLOS ONE*, *10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Salih, A., Raisi-Estabragh, Z., Galazzo, I. B., Radeva, P., Petersen, S. E., Menegaz, G., & Lekadir, K. (2023). *Commentary on explainable artificial intelligence methods: SHAP and LIME*. ArXiv.org. https://doi.org/10.48550/arXiv.2305.02012

Shafti, M., Taylor, P., Forrester, A., Handerer, F., & Pratt, D. (2023). A systematic review of the co-occurrence of self-harm and aggression: Is dual harm a unique behavioural construct? *Frontiers in Psychiatry*, *14*. https://doi.org/10.3389/fpsyt.2023.1083271

Silva, E. (2020). The HCR-20 and violence risk assessment – will a peak of inflated expectations turn to a trough of disillusionment? *BJPsych Bulletin*, *44*(6), 1–3. https://doi.org/10.1192/bjb.2020.14

Song, H., Rajan, D., Thiagarajan, J. J., & Spanias, A. (2017). Attend and Diagnose: Clinical Time Series Analysis using Attention Models. *ArXiv:1711.03905 [Cs, Stat]*. https://arxiv.org/abs/1711.03905

St Andrews Healthcare. (2025). *Seclusion Policy v2.0 Jul 25*. Sharepoint.com. ***[link redacted]***.

Suchting, R., Green, C. E., Glazier, S., & Lane, S. (2018). *A data science approach to predicting patient aggressive events in a psychiatric hospital*. Science Direct; Psychiatry Research. https://www.sciencedirect.com/science/article/abs/pii/S0165178118302853

Sugarman, P., Walker, L., & St Andrews Healthcare. (2007). *Health of the Nation Outcome Scales (HoNOS) for Users of Secure and Forensic Services HoNOS-secure: Security Scales A to G*. Royal College of Psychiatrists.

https://www.rcpsych.ac.uk/docs/default-source/improving-care/ccqi/honos/honos-secure-glossary-v2b-feb-07.pdf?sfvrsn=46fd0567_2

UK Census 2021. (2021). *How life has changed in West Northamptonshire: Census 2021*. Ons.gov.uk. https://www.ons.gov.uk/visualisations/censusareachanges/E06000062/

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017, June 12). *Attention Is All You Need*. ArXiv. https://arxiv.org/abs/1706.03762.

Transformers fundamentals.

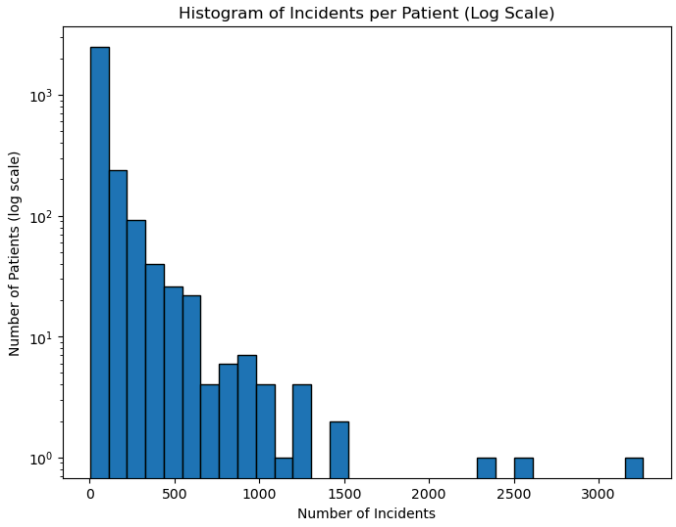
Vickers, A. J., & Elkin, E. B. (2006). Decision Curve Analysis: A Novel Method for Evaluating Prediction Models. *Medical Decision Making*, *26*(6), 565–574. https://doi.org/10.1177/0272989x06295361

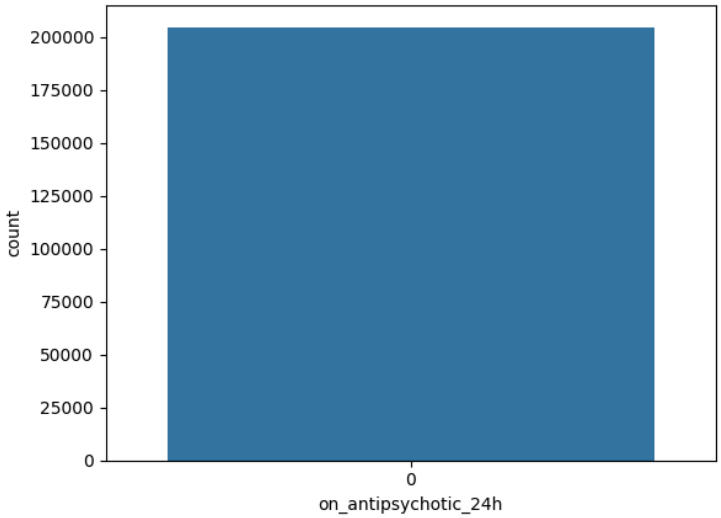
Woods, P., & Almvik, R. (2002). The Broset violence checklist (BVC). *Acta Psychiatrica Scandinavica*, *106*(s412), 103–105. https://doi.org/10.1034/j.1600-0447.106.s412.22.x

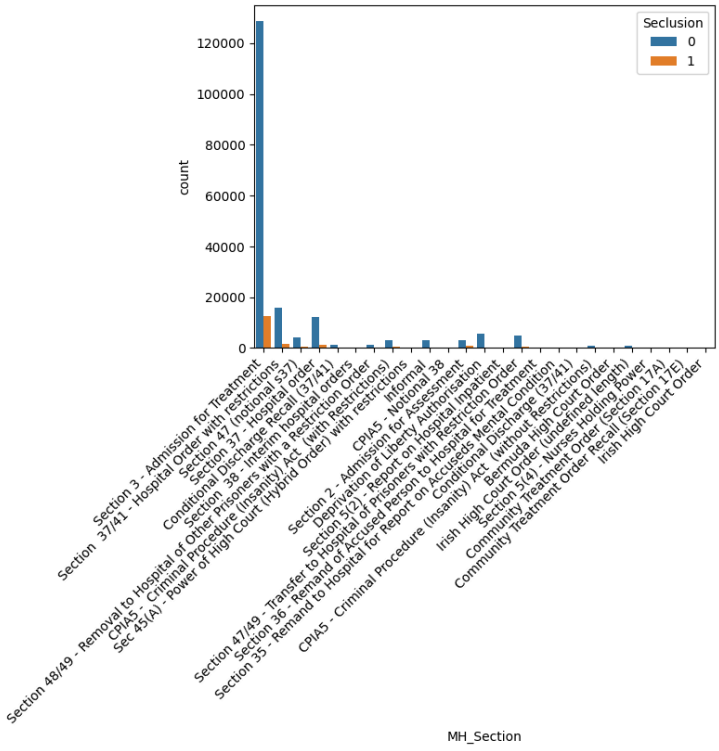
World Health Organization. (2019). *ICD-10*. ICD.WHO.INT. https://icd.who.int/browse10/2019/en

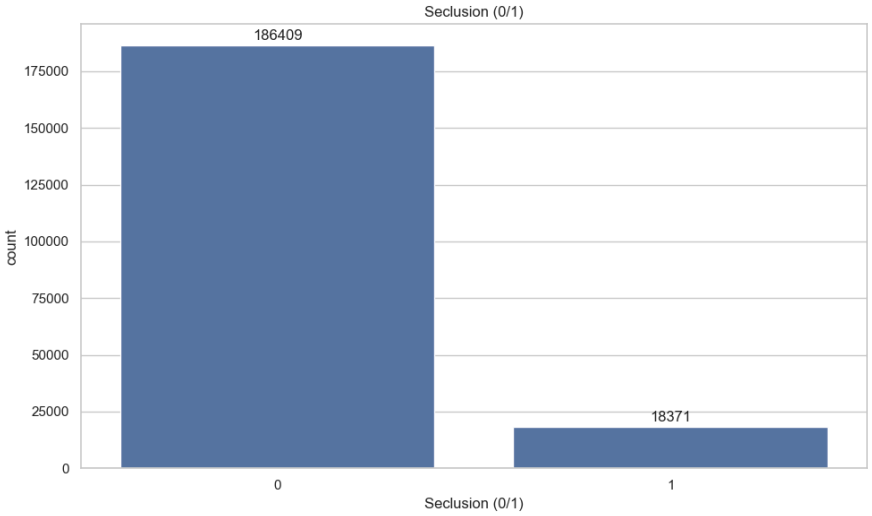
# 9 Appendix

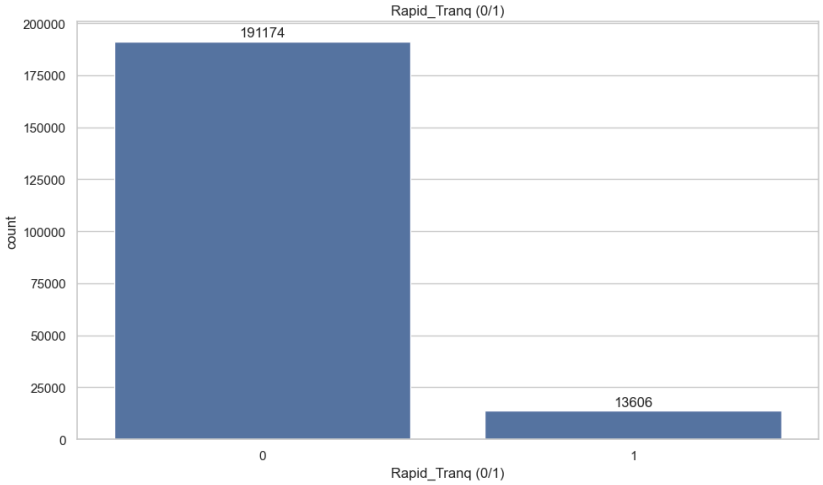
*Additional unused figures;*

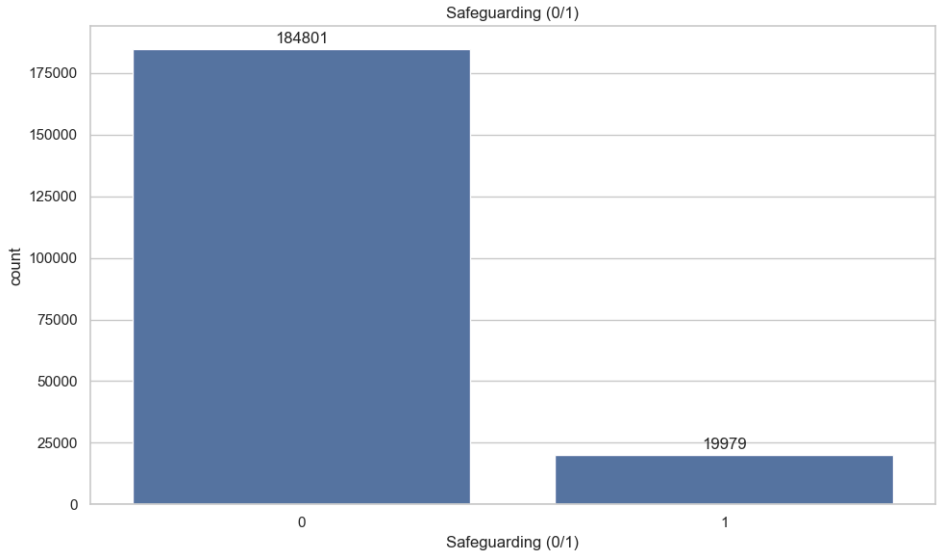


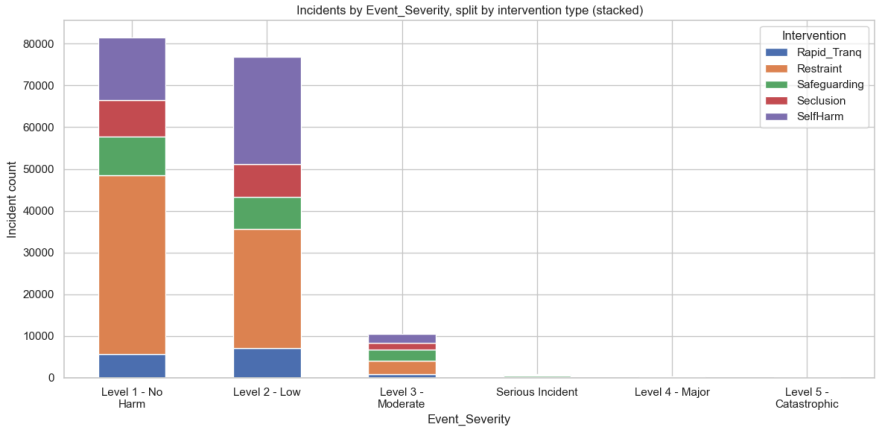

200000
175000
150000
125000
100000
75000
50000
25000
0
count
0
on_antipsychotic_24h

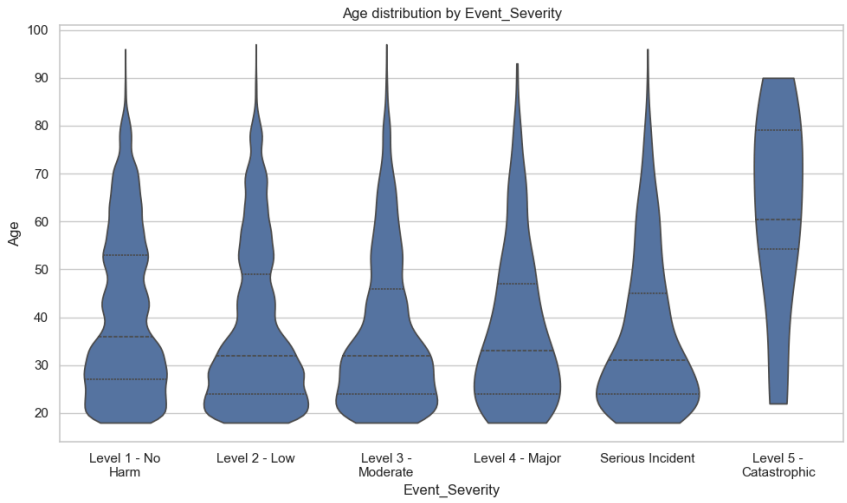

Seclusion
0
1
120000
100000
80000
60000
40000
20000
0
count
Section 3 - Admission for Treatment
Section 37/41 - Hospital Order with restrictions
Section 47 (notional s37)
Section 37 - Hospital order
Conditional Discharge Recall (37/41)
Section 38 - Interim hospital orders
Other Prisoners with a Restriction Order
Section 48/49 - Removal to Hospital of Other Prisoners with a Restriction Order
CPIA5 - Criminal Procedure (Insanity) Act (with Restrictions)
Sec 45(A) - Power of High Court (Hybrid Order) with restrictions
Informal
CPIA5 - Notional 38
Section 2 - Admission for Assessment
Deprivation of Liberty Authorisation
Section 5(2) - Report on Hospital Inpatient
Transfer to Hospital of Accused Prisoners with Restriction Order
Section 47/49 - Remand of Accused Person to Hospital for Treatment
Section 36 - Remand to Hospital for Report on Accused's Mental Condition
Section 35 - Conditional Discharge (37/41)
CPIA5 - Criminal Procedure (Insanity) Act (without Restrictions)
Bermuda High Court Order
Irish High Court Order (undefined length)
Section 5(4) - Nurses Holding Power
Community Treatment Order (Section 17A)
Community Treatment Order (Section 17E)
Community Treatment Order Recall (Section 17E)
Irish High Court Order
MH_Section

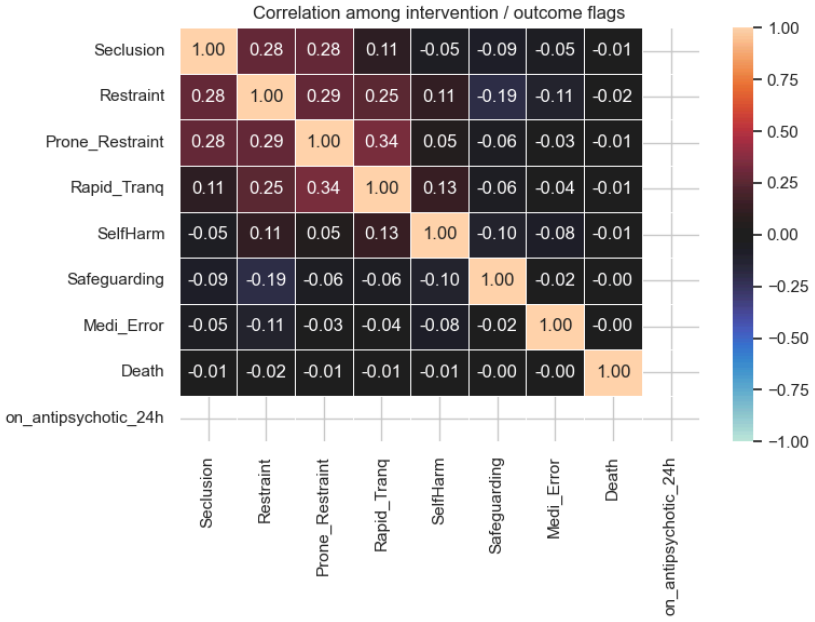

Seclusion (0/1)
186409
18371
count
0
25000
50000
75000
100000
125000
150000
175000
0
1
Seclusion (0/1)


Rapid_Tranq (0/1)
191174
13606
count
0
25000
50000
75000
100000
125000
150000
175000
200000
0
1
Rapid_Tranq (0/1)


Safeguarding (0/1)
184801
19979
count
0
25000
50000
75000
100000
125000
150000
175000
0
1
Safeguarding (0/1)


Incidents by Event_Severity, split by intervention type (stacked)
Intervention
Rapid_Tranq
Restraint
Safeguarding
Seclusion
Self-Harm
Incident count
0
10000
20000
30000
40000
50000
60000
70000
80000
Level 1 - No Harm
Level 2 - Low
Level 3 - Moderate
Serious Incident
Level 4 - Major
Level 5 - Catastrophic
Event_Severity


Age distribution by Event_Severity
Age
20
30
40
50
60
70
80
90
100
Level 1 - No Harm
Level 2 - Low
Level 3 - Moderate
Level 4 - Major
Serious Incident
Level 5 - Catastrophic
Event_Severity


Correlation among intervention / outcome flags
Seclusion 1.00 0.28 0.28 0.11 -0.05 -0.09 -0.05 -0.01
Restraint 0.28 1.00 0.29 0.25 0.11 -0.19 -0.11 -0.02
Prone_Restraint 0.28 0.29 1.00 0.34 0.05 -0.06 -0.03 -0.01
Rapid_Tranq 0.11 0.25 0.34 1.00 0.13 -0.06 -0.04 -0.01
SelfHarm -0.05 0.11 0.05 0.13 1.00 -0.10 -0.08 -0.01
Safeguarding -0.09 -0.19 -0.06 -0.06 -0.10 1.00 -0.02 -0.00
Medi_Error -0.05 -0.11 -0.03 -0.04 -0.08 -0.02 1.00 -0.00
Death -0.01 -0.02 -0.01 -0.01 -0.01 -0.00 -0.00 1.00
on_antipsychotic_24h
Seclusion
Restraint
Prone_Restraint
Rapid_Tranq
SelfHarm
Safeguarding
Medi_Error
Death
on_antipsychotic_24h
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00